AI ENGINEER

MASTER SYSTEMS + INTERVIEW PLAYBOOK

AI Engineering + Agentic AI + Data Science + Data Engineering

Hinglish-first rule

Concept English technical terms ke saath explain honge, lekin explanation, interview answers, examples aur practice instructions Hinglish me rahenge. Core coding labs me JavaScript/TypeScript, SQL, Bash, JSON, YAML aur pseudo-code use kiya gaya hai.

01 **Learn** Concepts	02 **Interview** Q&A + Mock	03 **Coding Lab** Build + Run
04 **Logic** Think + Solve	05 **Debug** Find + Fix	06 **Practice** Hands-on
07 **Mini Project** Build + Apply	08 **Defense** Explain + Present	09 **Checkpoint** Test Mastery

Source Base + 2026 Expansion

Uploaded sources ko consolidate karke 2026 framework/protocol verification ke saath expand kiya gaya hai. Technical names original rakhe gaye hain; explanations Roman Hinglish me hain.

Version 15 | Advanced+ | Project-ready | Interview-ready | AI + Data Systems

00 MASTER INDEX / PADHAI KA ROADMAP

Is index ko learning order me use karo. Har section me concept -> example -> code -> interview -> practice -> project link hai.

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01 ROLE MAP + AI ENGINEER SOCHNE KA TARIKA

AI Engineer ko sirf LLM API caller nahi, balki data + model + retrieval + agent + production system builder samjho.

Tumhe kaunse 4 roles ki knowledge combine karni hai?

Role	Core Skills	Mindset
AI Engineer	LLM apps, RAG, agents, tools, APIs, deployment, evals	System ko end-to-end reliable banana
Data Scientist	EDA, statistics, experiments, ML, metrics, business reasoning	Data se decision aur prediction banana
Data Analyst	SQL, KPI, funnel, cohort, dashboards, storytelling	Business questions ko measurable answers me convert karna
Data Engineer	ETL/ELT, warehouse/lakehouse, batch/streaming, quality, orchestration	Reliable data supply chain banana

Golden mental model

Business problem -> data contract -> data pipeline -> model/LLM -> retrieval/tool layer -> orchestration -> API/UI -> evaluation -> monitoring -> feedback loop.

AI Engineer interview me kaise sochna hai

1. Problem ko measurable outcome me define karo.
2. Data/knowledge source aur freshness identify karo.
3. Deterministic logic aur LLM-driven steps alag karo.
4. Latency, quality, cost, safety aur failure modes pehle define karo.
5. Evaluation dataset bina production launch mat assume karo.
6. Human approval aur rollback path high-impact actions ke liye rakho.

Core architecture cheat sheet

```
User
 |
API / Auth / Rate Limit
 |
Orchestrator
 +--> LLM
 +--> Retriever -> Vector / SQL / Search
 +--> Tools -> CRM / DB / Web / Internal APIs
 +--> Memory / State
 |
Guardrails -> Eval -> Observability
 |
Response / Action
```

02 LLM KI BUNIYAAD + TRANSFORMER DEEP DIVE

Source material ke LLM history, tokenization, embeddings, transformer, generation aur limitations ko interview-depth tak expand kiya gaya hai.

LLM ka core idea

LLM ek large neural model hai jo token sequences par patterns learn karta hai. Source material me isse next-token prediction, large text corpora, transformer architecture aur downstream adaptation ke context me explain kiya gaya hai.

Transformer pipeline

7. Text -> tokenization -> token IDs
8. Token IDs -> embeddings + positional information
9. Self-attention -> context mixing
10. Feed-forward / MLP blocks -> nonlinear transformation
11. Residual connections + normalization -> stable deep network
12. Output head -> token probability distribution
13. Decoding -> next token selection until stop condition

Attention intuition

Q = query, K = key, V = value. Attention ka goal hai current token ke liye useful context ko weighted combination ke form me lana. Common scaled dot-product form: softmax(QK^T / sqrt(dk))V. Interview me formula ke saath shape intuition bhi explain karo.

Scaled dot-product attention [PSEUDO]

```
scores = (Q @ K.T) / sqrt(d_k)
weights = softmax(scores)
context = weights @ V
```

Generation strategies

- Greedy: har step par highest-probability token.
- Beam search: multiple candidate sequences maintain karta hai.
- Top-k: fixed k likely tokens se sample.
- Top-p: cumulative probability threshold tak candidates.
- Temperature: distribution ko sharper ya flatter banata hai.

LLM limitations - production angle

- Hallucination: plausible but unsupported output.
- Freshness gap: model weights external live state nahi hote.
- Context overload: long context quality automatically perfect nahi hoti.
- Tool risk: tool-call arguments unsafe ya incorrect ho sakte hain.
- Prompt injection: retrieved/user content instructions ko hijack kar sakta hai.
- Cost/latency: large context aur multi-step agent loops expensive ho sakte hain.

Interview line

"LLM ko knowledge source nahi, probabilistic reasoning/generation component samajhta hoon. Grounded facts ke liye retrieval/tool layer aur correctness ke liye evaluation + guardrails use karunga."

03 PROMPT + CONTEXT ENGINEERING + STRUCTURED OUTPUT

Prompt ko sirf prompt nahi; instruction hierarchy, context selection, schemas, tool policy aur evaluation ka contract samjho.

Prompt engineering vs context engineering

Prompt engineering me instructions/examples improve karte ho. Context engineering me tum decide karte ho ki model ko exactly kaunsa state, history, retrieved evidence, tool result aur metadata diya jaye.

Production prompt skeleton

```
SYSTEM
Role: customer-support agent
Goal: solve request using approved tools
Rules:
- Never invent account facts
- Use tool for live status
- Ask before irreversible action
Output schema: JSON

CONTEXT
- customer_profile
- retrieved_policies
- current_ticket

USER
{{request}}
```

Structured output checklist

- Schema strict rakho.
- Enum values define karo.
- Required fields distinguish karo.
- Tool arguments ko separately validate karo.
- Raw model text ko direct SQL/command me inject mat karo.
- Invalid output -> retry/repair -> safe fallback.

Schema idea [TS]

```
const Action = z.object({
 action: z.enum(["answer", "search", "refund_request"]),
 reason: z.string(),
 needsApproval: z.boolean()
});
```

Advanced prompt interview topics

Sawal - Zero-shot aur few-shot me difference?

Jawaab - Zero-shot me example nahi hota. Few-shot me selected examples model ko desired pattern/format samjhate hain.

Level: Basic -> Advanced

Sawal - System/user/tool messages ko kaise design karoge?

Jawaab - Trusted policies ko high-priority instructions me rakho; user content ko data treat karo; tool outputs ko untrusted external data ki tarah validate karo.

Level: Basic -> Advanced

Sawal - Prompt injection ko kaise handle karoge?

Jawaab - Instruction/data boundary define karo, retrieved text ko non-authoritative mark karo, tool permissions narrow rakho, sensitive actions par approval gate lagao aur adversarial tests chalao.

Level: Basic -> Advanced

Sawal - Long context ka simple solution kya hai?

Jawaab - Sab kuch dump karna nahi; retrieval, compression, summarization, recency/priority ranking aur task-specific context windows use karo.

Level: Basic -> Advanced

Sawal - Context budget kaise decide karoge?

Jawaab - Har request ke liye input, retrieved evidence, tool results aur output ke liye budget define karunga; low-value history ko compress/drop karunga.

Level: Basic -> Advanced

Sawal - Prompt me business rules kahan rakhoge?

Jawaab - Trusted business rules ko controlled system/policy layer me rakho; user-provided text ko authoritative instruction nahi maanunga.

Level: Basic -> Advanced

Sawal - Structured output sirf JSON hona enough hai?

Jawaab - Nahi. Schema validation, type constraints, allowed enums, business-rule validation aur side-effect checks bhi chahiye.

Level: Basic -> Advanced

Sawal - Prompt regression kaise test karoge?

Jawaab - Versioned eval set par old vs new prompt run karke correctness, safety, tool selection, latency aur cost compare karunga.

Level: Basic -> Advanced

Sawal - Tool result bahut bada ho to?

Jawaab - Relevant fields extract, summarize or paginate karo; raw giant payload ko directly model context me mat dump karo.

Level: Basic -> Advanced

04 RAG ARCHITECTURE - BASIC TO ADVANCED

Uploaded LLM sources me RAG ko retriever + generator, embeddings + vector store + retriever + LLM + optional reranker ke roop me explain kiya gaya hai; yahan usko production architecture tak badhaya gaya hai.

RAG ka full pipeline

```
INGEST
Documents -> parse -> clean -> split -> metadata -> embed -> index

QUERY
Question -> normalize -> query rewrite -> retrieve -> rerank -> context pack -> LLM -> citations -> validator
```

Retrieval layers

- Dense retrieval: semantic similarity.
- Sparse/keyword retrieval: exact terms and identifiers.
- Hybrid retrieval: dense + keyword.
- Reranking: candidate results ko stronger relevance model se reorder karna.
- Metadata filters: tenant, department, date, permission, document type.
- Multi-query / decomposition: hard question ko multiple retrieval intents me break karna.

Chunking strategy

- Fixed-size chunk: easy but semantic boundaries weak ho sakte hain.
- Recursive/structure-aware chunk: headings, paragraphs, sections preserve.
- Parent-child retrieval: small chunk retrieve, larger parent context pass.
- Overlap carefully use karo; unnecessary overlap index size badhata hai.
- Chunk ke saath title, source, ACL, timestamp, page, product ID jaise metadata preserve karo.

RAG evaluation

Do metrics ko separate rakho

Retrieval metrics: hit@k, recall@k, MRR/NDCG. Generation metrics: answer correctness, groundedness, citation correctness, refusal correctness. Retrieval fail ko generation prompt se hide mat karo.

Advanced RAG failure matrix

Failure	Fix pattern
Wrong chunk	better chunking + hybrid retrieval + reranker
Right chunk but wrong answer	grounding prompt + citation requirement + answer verifier
Stale info	freshness metadata + re-indexing + source TTL
Unauthorized data leakage	ACL-aware retrieval + tenant filter + post-retrieval authorization
Slow retrieval	ANN index + caching + fewer candidates + parallel retrieval

05 LANGCHAIN: MODERN AGENT HARNESS

Current official docs ke according LangChain me `create_agent` ek minimal configurable agent harness hai; tools, prompt aur middleware ko compose kiya jata hai. LangChain agents LangGraph ke upar built hain.

LangChain ko kab use karna hai?

- Fast agent/tool-calling application.
- Provider-agnostic model + tool interfaces.
- Middleware based guardrails, retries, routing, policies.
- Simple-to-advanced RAG composition.
- LangGraph use karo jab explicit stateful orchestration aur durable control chahiye.

Modern JS/TS agent skeleton

Current-style LangChain agent [TS]

```
import { createAgent, tool } from "langchain";
import * as z from "zod";

const getOrder = tool(
 async ({ orderId }) => ({ orderId, status: "SHIPPED" }),
 {
  name: "get_order",
  description: "Get live order status",
  schema: z.object({ orderId: z.string() })
 }
);

const agent = createAgent({
 model: "provider:model-id",
 tools: [getOrder]
});

await agent.invoke({
 messages: [{ role: "user", content: "Order AB123 ka status?" }]
});
```

LangChain concepts jo interview me aane chahiye

- Model interface
- Tool schema
- Middleware
- Agent loop
- Message/state handling
- Retrievers and vector stores
- Prompt templates
- Structured output
- LangSmith tracing/evaluation
- LangGraph integration

06 LANGGRAPH: STATEFUL ORCHESTRATION

Current docs LangGraph ko low-level orchestration runtime batati hain jo long-running, stateful agents ke liye durable execution, streaming, human-in-the-loop aur persistence provide karta hai.

LangChain vs LangGraph

LangChain = agent framework/harness. LangGraph = orchestration runtime. Simple tool-calling flow ke liye LangChain enough ho sakta hai; complex state machine, branching, retries, approvals aur long-running workflow me LangGraph explicit control deta hai.

Graph thinking

```
START
 -> classify
   -> [FAQ] answer
   -> [ORDER] get_order
   -> [REFUND] human_approval
 -> validate
 -> END
```

Graph skeleton [TS]

```
import { StateGraph, START, END } from "@langchain/langgraph";

// Node = ek deterministic ya agentic step
// Edge = next transition
// State = shared workflow data

const graph = new StateGraph(State)
 .addNode("classify", classify)
 .addNode("answer", answer)
 .addNode("action", action)
 .addEdge(START, "classify")
 // conditional routing here
 .compile();
```

State design rules

- State me sirf needed fields rakho.
- User identity aur authorization state ko explicit rakho.
- Intermediate tool outputs ko typed structure do.
- Long-running tasks ke liye resumable checkpoints design karo.
- Human approval step ko graph state me first-class state banao.

07 AGENTIC AI: AGENTS, PLANNING, TOOLS, MEMORY

Agent = model + state + tools + policy + loop. Uploaded AVO research autonomous planning, implementation, testing, debugging aur persistent memory based loop ko variation operator ke roop me describe karta hai.

Agent anatomy

```
GOAL
 -> PLAN
 -> SELECT TOOL / ACTION
 -> EXECUTE
 -> OBSERVE RESULT
 -> VERIFY
 -> REPLAN or COMPLETE
```

Agent types

- Reactive agent: current input -> action.
- Tool-using agent: model decides which tool to call.
- Planner-executor: plan first, actions later.
- ReAct-style loop: reason/action/observation pattern.
- Workflow agent: deterministic graph + selective LLM decisions.
- Multi-agent: specialized agents with explicit handoffs.
- Autonomous optimization agent: long-running search with evaluation feedback, as in AVO.

Agent memory

- Working memory: current task state.
- Conversation memory: user interaction history.
- Long-term memory: durable user/project facts, subject to privacy and authorization.
- Episodic memory: past successful/failed episodes.
- Semantic memory: structured facts or indexed knowledge.

Tool design - important rule

Least privilege

Agent ko "database access" mat do; narrow tool do, e.g. `get_customer_order(orderId)` with strict schema. Write/destructive tools alag permission class me rakho aur human approval ya policy gate lagao.

AVO-inspired autonomous improvement

Uploaded AVO paper ke diagram aur text me agent ko previous solutions, tools, memory, domain knowledge aur evaluation utility ke saath iterative planning/implementation/testing/debugging loop me rakha gaya hai. Is pattern ko coding-agent, optimizer aur engineering workflow projects me adapt kiya ja sakta hai.

08 MCP: TOOLS AUR CONTEXT CONNECTIVITY

MCP ko agent-tool connectivity standard ki tarah samjho. July 28, 2026 spec ne stateless protocol core, header-based routing, cacheable list results aur authorization hardening add kiye.

MCP basic model

```
HOST / AGENT
  |
 MCP CLIENT
  |
-------------------
| MCP SERVER A    | -> tools / data
| MCP SERVER B    | -> resources / prompts
-------------------
```

Important primitives

- Tools: model-invoked actions such as search, database query, API call. Official MCP tool spec says tools can interact with external systems and have input schema metadata.
- Resources: contextual data exposed by server.
- Prompts: reusable prompt templates / interaction patterns.
- Capabilities: client/server feature negotiation.
- Authorization: identity, scopes, validation and user consent matter in remote deployments.

2026 MCP interview points

Sawal - 2026-07-28 MCP release ka big architectural change kya hai?

Jawaab - Protocol core stateless banaya gaya; requests self-describing ho sakte hain, aur normal load balancer ke peeche instances route kiye ja sakte hain.

Level: Advanced

Sawal - MCP tool ko secure kaise rakhoge?

Jawaab - Narrow schema, least privilege, explicit authorization, output validation, audit logs, rate limits, user approval for sensitive actions, aur untrusted server content ko trusted instructions na maana.

Level: Advanced

MCP tool-call data flow

Simplified current-spec shape [HTTP/JSON]

```
POST /mcp
MCP-Protocol-Version: 2026-07-28
Mcp-Method: tools/call
Mcp-Name: search

{
 "jsonrpc": "2.0",
 "id": 1,
 "method": "tools/call",
 "params": {
  "name": "search",
  "arguments": {"q": "AI engineer"}
 }
}
```

09 MULTI-AGENT + DEEP AGENTS + AUTONOMOUS WORKFLOWS

Specialization, handoff, state, verification aur cost control ke bina multi-agent system complexity badha sakta hai.

Patterns

- Supervisor -> specialists -> verifier
- Router -> domain agents
- Parallel specialists -> aggregator
- Debate/critique -> finalizer
- Planner -> executor -> tester
- Background agent -> human approval -> action

When NOT to use multi-agent

- Task simple deterministic hai.
- Ek agent + 2-3 tools sufficient hai.
- Latency/cost budget tight hai.
- Evaluation weak hai.
- Shared state/ownership unclear hai.

Deep-agent feature checklist

- Planning
- Subagent spawning
- Filesystem/tool access
- Context compression
- Long-running task support
- Resume from checkpoint
- Human approval
- Observability

Agentic workflow example

```
REQUEST
 -> Intake Agent
 -> Research Agent ----+
 -> Data Agent --------+--> Verifier --> Writer --> Human Approval
 -> Tool Agent --------+
```

10 DATA SCIENCE: ANALYTICS + STATISTICS + ML

Uploaded curriculum me data basics, visualization, probability, confidence intervals, hypothesis testing, regression, clustering, ML/DL, NLP aur GenAI use cases included hain.

Data analysis lifecycle

14. Business question
15. Data collection
16. Data quality checks
17. EDA
18. Hypothesis / metric design
19. Modeling or analysis
20. Validation
21. Visualization/storytelling
22. Decision + monitoring

Statistics jo AI Engineer ko bhi aani chahiye

- Mean, median, mode, variance, standard deviation
- Percentile, quartile, IQR, outliers
- Probability and conditional probability
- Random variables and distributions
- Confidence intervals
- p-value, significance level
- Type I / Type II error
- t-tests and chi-square
- Correlation vs causation
- Regression

ML fundamentals

- Supervised / unsupervised / reinforcement learning
- Regression / classification / clustering
- Train/validation/test split
- Cross-validation
- Overfitting/underfitting
- Bias/variance
- Feature engineering
- Calibration and thresholding
- Precision/recall/F1/ROC-AUC/PR-AUC
- Business metric alignment

Data Analyst interview examples

Sawal - Dashboard me 20 KPIs dikhana sahi hai?

Jawaab - Usually nahi. User role aur decision ke hisaab se top KPIs, drill-down, filters aur alerts design karo. Dashboard ka purpose decision enable karna hai, data dump karna nahi.

Level: Basic

Sawal - Correlation high ho to causation prove hota hai?

Jawaab - Nahi. Confounding, reverse causality aur selection bias possible hain. Causal claim ke liye experiment ya stronger identification strategy chahiye.

Level: Normal

Sawal - Class imbalance me accuracy misleading kyun ho sakti hai?

Jawaab - Agar positive class rare hai to all-negative prediction bhi high accuracy de sakta hai. Precision, recall, PR-AUC aur business cost matrix zyada informative ho sakte hain.

Level: Normal

Sawal - Model probability calibrated kyun hona chahiye?

Jawaab - Agar 0.8 score ka matlab repeatedly 80% event likelihood nahi hai, to thresholding aur risk decisions unreliable ho sakte hain. Calibration curves/Brier-style checks useful hain.

Level: Advanced

Sawal - Train-test split me group leakage kya hai?

Jawaab - Same customer/device/document ke related rows train aur test dono me chale gaye to score unrealistically high ho sakta hai. Group-aware split use karo.

Level: Advanced

Sawal - Experiment me primary aur guardrail metric kya?

Jawaab - Primary metric main success criterion hai; guardrail metric harmful side-effects jaise latency, churn, refunds ya error rate detect karta hai.

Level: Normal

Sawal - Feature distribution production me badal jaye to?

Jawaab - Drift monitor karo, root cause dekho, threshold/model retrain ya feature pipeline fix karo, aur historical comparison maintain karo.

Level: Advanced

Source curriculum se practical use cases

- Customer segmentation using clustering
- Sales forecasting
- Customer feedback/sentiment analysis
- Product recommendations
- AI-generated business reports
- Multilingual summarization
- Document classification/tagging

11 DATA ENGINEERING - WAREHOUSE, LAKEHOUSE, STREAMING, QUALITY

Uploaded curriculum me Hadoop, Spark, Kafka, Big Data, scalability aur streaming analytics included hain.

Modern data platform mental model

```
Sources
  |
CDC / Events / Files / APIs
  |
Ingestion -> Kafka / Object Storage
  |
Bronze -> Silver -> Gold
  |
Lakehouse / Warehouse
  |
BI / DS / ML / AI / RAG
```

Batch vs streaming

- Batch: periodic bounded processing.
- Streaming: continuous event processing.
- Micro-batch: small frequent batches.
- Event-time vs processing-time: event kab hua vs system ne kab process kiya.
- Watermarks: late events handle karne ke liye state bounds.
- Idempotency: retry ke baad duplicate business effect avoid karna.

Data modeling

- OLTP: normalized, transaction-focused.
- OLAP/warehouse: analytical queries.
- Star schema: fact + dimensions.
- Slowly Changing Dimensions: historical dimension changes track karna.
- Data contracts: schema + semantics + ownership + SLA.
- Partitioning: pruning aur scalability.
- Clustering/sorting: scan efficiency.

Lakehouse concepts

- ACID transactions on object-store tables.
- Schema evolution.
- Time travel/versioning.
- Streaming + batch on same table.
- Upsert/merge for CDC.
- Partition evolution / hidden partitioning depending on table format.

Delta Lake docs ACID transactions, schema enforcement, time travel, streaming/batch unification aur upserts/deletes ko lakehouse use cases ke liye describe karti hain. Apache Iceberg docs schema evolution, hidden partitioning, time travel aur rollback ko highlight karti hain.

Kafka concepts

- Topic
- Partition
- Offset
- Producer
- Consumer

- Consumer group
- Key-based partitioning
- Ordering per partition
- Retention
- Replay

Spark concepts

- Distributed DataFrame/SQL processing
- Partitions and shuffles
- Joins and skew
- Caching/persistence
- Structured Streaming
- Exactly-once vs at-least-once semantics in end-to-end design

Spark Structured Streaming official guide current stream-processing capabilities ka reference hai.

Orchestration - Airflow

- DAG = workflow definition.
- Task = execution unit.
- Dependencies = order.
- Scheduler = ready tasks trigger karta hai.
- Retry/timeout/backfill/concurrency important operational concepts hain.
- Airflow docs ke current concept pages DAGs, tasks, scheduler, workers aur dependencies ko core workflow concepts batate hain.

Data quality

- Freshness
- Completeness
- Validity
- Uniqueness
- Consistency
- Referential integrity
- Schema conformance
- Business-rule checks

Great Expectations current docs Expectations ko verifiable assertions ke roop me describe karti hain aur expectation suites/validation workflow ko data quality checks ke liye use karti hain.

12 MLOPS + LLMOPS + EVALUATION + OBSERVABILITY

Model quality se zyada important hai production behavior ko measurable banana.

MLOps lifecycle

```
Data -> Train -> Validate -> Register -> Deploy -> Monitor
      ^                                      |
      |---- Retrain / rollback -|
```

LLMOps stack

- Prompt/version management
- Model/version routing
- Dataset + eval set versioning
- Trace each request/tool call
- Latency and token usage
- Quality metrics
- Safety tests
- Cost budgets
- Rollback

Evaluation dimensions

- Task correctness
- Groundedness
- Citation correctness
- Tool-call correctness
- Policy compliance
- Refusal correctness
- Latency
- Cost
- Robustness
- Human preference / expert review

Observability

```
trace_id
 span: request
 span: retrieval
 span: llm_call
 span: tool_call
 span: verifier
 metrics: latency, tokens, cost, errors, quality
```

LangChain docs LangSmith ko tracing, debugging aur evaluation ke liye position karti hain; LangGraph docs execution paths, state transitions aur runtime metrics ke visibility use cases mention karti hain.

Drift

- Data drift: inputs distribution changes.
- Concept drift: input-output relationship changes.
- Model drift: performance/behavior degrades.
- LLM app drift: prompt, tool, corpus, model/provider or user behavior changes.

13 AI SECURITY + GOVERNANCE + RESPONSIBLE AI

Production AI me security sirf API key protection nahi: prompt injection, tool abuse, data leakage, supply chain, authorization aur evaluation sab cover karo.

Top security controls

- Strong authn/authz
- Tenant isolation
- Least-privilege tools
- Input/output validation
- Prompt injection defenses
- Secrets management
- PII redaction/minimization
- Audit logs
- Rate limits and quotas
- Sandbox for code execution
- Human approval for high-impact actions
- Red-team tests

Threat -> control examples

Threat	Control
Prompt injection	Instruction/data separation + retrieval sanitization + tool policy
Sensitive data leakage	ACL filters + output checks + redaction
Excessive agency	Narrow tools + approval gates + rate limits
Tool injection / malicious tool metadata	Server trust controls + schema validation + allowlists
Supply chain risk	Pinned dependencies + signatures/SBOM + provenance
Model misuse	Policy layer + monitoring + abuse testing

NIST AI RMF ka purpose AI risks ko manage karna aur trustworthiness considerations ko design, development, use aur evaluation me include karna hai; NIST ne GenAI Profile bhi publish kiya hai.

14 AI SYSTEM DESIGN + SCALABILITY + COST

Interview me ek architecture diagram ke saath trade-offs explain karne ki habit banao.

Scalable AI chat architecture

```
Client
 -> API Gateway
 -> Auth / Rate Limit
 -> Request Router
 -> Agent Orchestrator
  +-- Retrieval Service
  +-- Tool Gateway
  +-- Model Router
  +-- Memory Store
 -> Response Validator
 -> Stream to Client

Infra: cache + queue + object store + DB + observability
```

Key trade-offs

- Latency vs quality
- Cost vs model capability
- Context size vs retrieval precision
- Synchronous vs asynchronous tools
- Single agent vs workflow graph vs multi-agent
- Managed service vs self-hosted model
- Strong consistency vs throughput
- Online serving vs batch processing

Reliability patterns

- Timeouts
- Retries with backoff
- Circuit breaker
- Idempotency keys
- Dead-letter queue
- Fallback model
- Graceful degradation
- Bulkheads
- Caching
- Health/readiness checks

Cost controls

- Route easy queries to smaller/cheaper models.
- Cache stable retrieval and prompts.
- Limit top-k and context size.
- Batch offline workloads.
- Budget per user/tenant/workflow.
- Track tokens and tool costs by trace.

Async agent task

Long-running tasks ko synchronous HTTP request me force mat karo. Job ID + status store + background worker + webhook/polling + cancellation/timeout pattern use karo. MCP 2026 Tasks extension draft server-directed asynchronous tasks ka model describe karti hai.

15 CODING LABS: PRACTICAL IMPLEMENTATION

Ye labs AI Engineer + Data Analyst + Data Engineer work ke practical patterns cover karte hain.

Lab 01 - Tool schema validation

```
const toolInput = z.object({
 customerId: z.string().min(1),
 action: z.enum(["lookup", "update"]),
 approvalId: z.string().optional()
});
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 02 - Safe API wrapper

```
const res = await fetch(url, {
 headers: { Authorization: `Bearer ${token}` }
});
if (!res.ok) throw new Error(`Upstream ${res.status}`);
const data = await res.json();
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 03 - LangChain agent

```
const agent = createAgent({
 model: "provider:model-id",
 tools: [searchTool, getTicketTool],
});
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 04 - RAG query contract

```
{
 "query": "refund policy",
 "filters": {"tenant":"acme","locale":"en-IN"},
 "topK": 8,
 "rerank": true
}
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 05 - SQL funnel

```
SELECT stage, COUNT(DISTINCT user_id) AS users
FROM funnel_events
GROUP BY stage
ORDER BY users DESC;
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 06 - Cohort retention

```
SELECT cohort_month, months_since_signup,
    COUNT(DISTINCT user_id) AS active_users
FROM retention
GROUP BY cohort_month, months_since_signup;
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 07 - Data quality check

```
SELECT
 COUNT(*) AS total_rows,
 COUNT(*) FILTER (WHERE customer_id IS NULL) AS null_customer_ids
FROM orders;
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 08 - CDC upsert idea

```
MERGE INTO dim_customer t
USING staging_customer s
ON t.customer_id = s.customer_id
WHEN MATCHED THEN UPDATE SET name=s.name, updated_at=s.updated_at
WHEN NOT MATCHED THEN INSERT (customer_id,name,updated_at) VALUES (s.customer_id,s.name,s.updated_at);
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 09 - Spark SQL transformation

```
SELECT customer_id, DATE_TRUNC("month", event_time) AS month,
    SUM(amount) AS revenue
FROM events
GROUP BY customer_id, DATE_TRUNC("month", event_time);
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 10 - Docker healthcheck

```
FROM node:22-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
CMD ["node","server.js"]
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 11 - Kubernetes deployment skeleton

```
apiVersion: apps/v1
kind: Deployment
spec:
 replicas: 3
 template:
  spec:
   containers:
    - name: ai-api
     image: registry/ai-api:1.0.0
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

Lab 12 - Evaluation case

```
{
 "input": "What is refund window?",
 "expected": "30 days",
 "mustCite": true,
 "mustUseTool": false
}
```

Practice: is lab ko apne project ke ek real use-case se connect karo; error path aur test case bhi likho.

16 LOGIC + DSA FOR AI ENGINEERS

Uploaded Algorithms Notes me Big-O, trees, BST, graph traversals, Dijkstra, A*, DP, greedy, KMP, sorting/searching aur related algorithms cover hain.

AI Engineer ke liye DSA focus

- Big-O and space complexity
- Hash maps / sets
- Stacks / queues / heaps
- Binary search
- Trees / tries
- Graphs / BFS / DFS
- Shortest paths
- Topological sort
- Dynamic programming
- Greedy patterns
- String search (KMP/Rabin-Karp)

Complexity quick table

Topic	Complexity	Interview note
Hash lookup avg	O(1)	caveat: collisions/load
Binary search	O(log n)	sorted/monotonic structure
Heap push/pop	O(log n)	peek O(1)
BFS/DFS	O(V+E)	adjacency list
Dijkstra	O((V+E) log V)	common heap implementation
Merge sort	O(n log n)	stable, extra space
DP LCS	O(nm)	state table

Logic pattern

23. Input constraints dekho.
24. Brute force identify karo.
25. Repeated work spot karo.
26. Data structure choose karo.
27. Time/space complexity calculate karo.
28. Edge cases likho.
29. Test with tiny + adversarial inputs.

17 MINI PROJECTS: END-TO-END BLUEPRINTS

Source capstone list me YouTube summarizer, multilingual summarizer, order bot, medical information agent, multi-agent support, job application tailor, personalization engine aur smart document classification included hain.

01 Enterprise RAG Support Agent

Architecture: Docs -> ingestion -> hybrid retrieval -> rerank -> LangChain agent -> escalation -> citations -> LangSmith eval

Key stack: RAG + agents + security + eval

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

02 Multi-Agent Customer Support

Architecture: Router -> order agent -> refund agent -> knowledge agent -> verifier -> human approval

Key stack: LangGraph + tools + HITL + memory

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

03 AI Job Application Tailor

Architecture: JD parser -> skill extractor -> resume matcher -> evidence validator -> document generator

Key stack: Structured output + retrieval + guardrails

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

04 Smart Document Intelligence

Architecture: OCR/parsing -> classify -> metadata -> extract entities -> validation -> index

Key stack: Multimodal + data quality + RAG

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

05 Data-to-Insight Copilot

Architecture: SQL tool -> semantic layer -> KPI checks -> chart spec -> narrative -> audit log

Key stack: Data Analyst + AI Engineer

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

06 Streaming Anomaly Agent

Architecture: Kafka events -> stream processor -> feature logic -> detector -> alert agent -> incident workflow

Key stack: Data Engineering + ML + agents

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

07 AVO-style Coding Optimizer

Architecture: Candidate -> benchmark -> profiler -> critique -> edit -> test -> persist -> repeat

Key stack: Agentic optimization + evaluation

- Must-have demo: happy path + one failure path
- Evaluation set: 20-100 labeled cases
- Security: auth + authorization + logging
- Defense: 90-second explanation + architecture diagram

18 INTERVIEW Q&A BANK - 270+ QUESTIONS

Har sawaal ka jawab Roman Hinglish me hai aur Basic -> Normal -> Advanced progression diya gaya hai. Answer ko interview me 30-60 sec me explain karne layak rakha gaya hai.

LLM & Transformers

Sawal - LLM kya hai?

Jawaab - Large Language Model large text/data corpora se language patterns learn karta hai aur token-by-token generation ya language understanding tasks perform karta hai.
Level: Basic

Sawal - Tokenization kya hai?

Jawaab - Text ko tokens me break karna. Model raw text ke bajay token IDs process karta hai.
Level: Basic

Sawal - Embedding kya hai?

Jawaab - Token/text ko dense numeric vector me map karna, jisse semantic relationships mathematically represent kiye ja sakte hain.
Level: Basic

Sawal - Transformer ki key strength kya hai?

Jawaab - Self-attention ki wajah se long-range relationships ko parallel processing ke saath model karna easy hota hai.
Level: Basic

Sawal - Self-attention ka intuition?

Jawaab - Current token ke liye sequence ke relevant tokens ko importance weights dekar context vector banana.
Level: Basic

Sawal - Q, K, V kya hain?

Jawaab - Query poochta hai mujhe kya relevant chahiye, Key batata hai token kis type ka context deta hai, Value actual information carry karta hai.
Level: Normal

Sawal - Positional information kyun chahiye?

Jawaab - Attention itself token order ko inherently encode nahi karta, isliye sequence position information add karni hoti hai.
Level: Normal

Sawal - Causal language modeling kya hai?

Jawaab - Previous tokens dekhkar next token predict karna.
Level: Normal

Sawal - Masked language modeling kya hai?

Jawaab - Sequence me kuch tokens hide karke model ko context se unhe predict karwana.
Level: Normal

Sawal - Temperature ka role?

Jawaab - Sampling distribution ki sharpness change karta hai; lower usually more deterministic, higher more diverse.
Level: Normal

Sawal - Top-k vs top-p?

Jawaab - Top-k fixed number of likely tokens rakhta hai; top-p cumulative probability threshold ke basis par variable candidate set banata hai.
Level: Advanced

Sawal - Context window kya hai?

Jawaab - Ek model request ke dauran available token budget jisme input/context/output constraints operate karte hain.
Level: Advanced

Sawal - Hallucination kya hai?

Jawaab - Model ka plausible-looking lekin unsupported/incorrect information generate karna.
Level: Advanced

Sawal - LLM hallucinate kyun karta hai?

Jawaab - Objective language continuation hai; truth verification guaranteed mechanism nahi hota jab external grounding absent ho.
Level: Advanced

Sawal - Fine-tuning kab use karoge?

Jawaab - Behavior/style/task adaptation ke liye jab prompt + retrieval se repeatable quality achieve na ho aur labeled training data available ho.
Level: Advanced

Sawal - LoRA kya hai?

Jawaab - Parameter-efficient fine-tuning approach jisme low-rank trainable updates use hote hain instead of all model parameters.
Level: Advanced

Sawal - Quantization kya hai?

Jawaab - Weights/activations ko lower precision representation me convert karke memory aur compute reduce karna.
Level: Advanced

Sawal - Distillation kya hai?

Jawaab - Large teacher model ke behavior ko smaller student model me transfer karna.
Level: Advanced

Sawal - Model selection kaise karoge?

Jawaab - Quality, latency, context, tool support, cost, privacy, reliability aur workload fit ke against benchmark karunga.
Level: Advanced

Sawal - LLM evaluation kyun hard hai?

Jawaab - Open-ended outputs me multiple correct answers ho sakte hain aur surface fluency correctness guarantee nahi karti.
Level: Advanced

Prompt + Context + Structured Output

Sawal - Prompt engineering kya hai?

Jawaab - Model ko effective instructions, constraints, examples aur output format dene ki practice.
Level: Basic

Sawal - Context engineering kya hai?

Jawaab - Task ke liye relevant state, retrieved evidence, history, tools aur metadata ko select/order/compress karna.
Level: Basic

Sawal - Few-shot prompting kab useful hai?

Jawaab - Jab exact output style, classification boundary ya schema behavior examples se clear karna ho.
Level: Basic

Sawal - Prompt injection kya hai?

Jawaab - Untrusted input prompt ke intended instructions ko override/manipulate karne ki koshish.
Level: Basic

Sawal - System aur user message ko kaise alag rakhoge?

Jawaab - Trusted policy/system instructions ko controlled layer me; user content ko data/input layer me treat karunga.
Level: Basic

Sawal - Structured output kyun important hai?

Jawaab - Downstream systems deterministic fields expect karte hain; schema validation automation reliability improve karta hai.
Level: Normal

Sawal - Invalid JSON aaye to?

Jawaab - Schema validation -> repair/retry -> safe fallback; invalid output ko blindly execute nahi karunga.
Level: Normal

Sawal - Long context ka best fix?

Jawaab - Relevant retrieval, compression, summarization, ranking aur context budgeting; blind full-history dump nahi.
Level: Normal

Sawal - Prompt versioning kyun?

Jawaab - Behavior regressions track, rollback aur A/B evaluation ke liye.
Level: Normal

Sawal - Prompt caching ka use?

Jawaab - Repeated static context/instructions ko reuse karke latency/cost reduce kar sakte hain, provider capability ke according.
Level: Normal

RAG

Sawal - RAG kya hai?

Jawaab - Retriever external knowledge fetch karta hai aur LLM retrieved context ke basis par answer generate karta hai.
Level: Basic

Sawal - RAG ki basic pipeline?

Jawaab - Ingest -> chunk -> embed -> index; query -> retrieve -> context -> generate.
Level: Basic

Sawal - Vector DB kya hai?

Jawaab - Embeddings ko efficiently store/search karne wala system.
Level: Basic

Sawal - Semantic search kya hai?

Jawaab - Meaning/similarity ke basis par relevant content find karna, exact keyword match se beyond.
Level: Basic

Sawal - Chunking ka impact?

Jawaab - Bad chunking retrieval quality reduce kar sakti hai; structure-aware chunking often better context boundaries preserve karti hai.
Level: Basic

Sawal - Hybrid search kya hai?

Jawaab - Keyword + semantic retrieval combine karna.
Level: Normal

Sawal - Reranker kyun?

Jawaab - Initial retriever ke candidate set ko better relevance model se reorder karne ke liye.
Level: Normal

Sawal - Metadata filter kyun?

Jawaab - Tenant, permission, date, product jaise constraints enforce karne ke liye.
Level: Normal

Sawal - RAG hallucination solve karta hai?

Jawaab - Automatically nahi. Good retrieval + grounded prompting + validation se unsupported answers reduce ho sakte hain.
Level: Normal

Sawal - RAG me citation ka role?

Jawaab - User ko answer ke supporting source evidence dikha sakte hain aur verification easy hota hai.
Level: Normal

Sawal - Multi-hop RAG kya hai?

Jawaab - Question ko multiple retrieval/answer steps me solve karna jahan second query first result par depend kare.
Level: Advanced

Sawal - Agentic RAG kya hai?

Jawaab - Agent retrieval strategy khud choose/iterate karta hai, multiple queries/tools use kar sakta hai aur verification kar sakta hai.
Level: Advanced

LangChain + LangGraph

Sawal - LangChain kya hai?

Jawaab - Models, tools aur agent loops ko compose karne ke liye framework/harness.
Level: Basic

Sawal - create_agent kya hai?

Jawaab - Current LangChain docs me configurable agent harness jo model, tools, prompt aur middleware combine karta hai.
Level: Basic

Sawal - Middleware ka use?

Jawaab - Guardrails, retries, routing, tool policies aur pre/post processing inject karne ke liye.
Level: Basic

Sawal - LangGraph kya hai?

Jawaab - Stateful, long-running agent workflows ke liye low-level orchestration runtime.
Level: Basic

Sawal - LangChain aur LangGraph difference?

Jawaab - LangChain higher-level agent abstractions deta hai; LangGraph explicit state graph/orchestration control deta hai.
Level: Basic

Sawal - LangGraph me state kya hai?

Jawaab - Workflow ke nodes ke across carry hone wala typed/shared data.
Level: Normal

Sawal - Persistence kya solve karti hai?

Jawaab - Workflow state checkpoints save karke failure ke baad resume aur long-running tasks enable karti hai.
Level: Normal

Sawal - HITL kya hai?

Jawaab - Human-in-the-loop: sensitive decision/action se pehle human inspection/approval.
Level: Normal

Sawal - Tool ko kaise define karoge?

Jawaab - Clear name, precise description, strict input schema, bounded side effects, auth context aur safe output.
Level: Normal

Sawal - LangSmith ka role?

Jawaab - Tracing, debugging aur evaluation ke liye visibility platform.
Level: Normal

Agents + Agentic AI

Sawal - Agent kya hai?

Jawaab - Model + state + tools + policy + execution loop jo goal ko multi-step actions me pursue karta hai.
Level: Basic

Sawal - Agent loop kya hota hai?

Jawaab - Plan/select action -> execute -> observe -> verify -> replan/finish.
Level: Basic

Sawal - Workflow aur agent me difference?

Jawaab - Workflow me transitions mostly predefined hote hain; agent dynamic decisions le sakta hai.
Level: Basic

Sawal - Planner-executor pattern?

Jawaab - Ek component plan banata hai, executor steps perform karta hai, verifier result check karta hai.
Level: Basic

Sawal - Single-agent vs multi-agent?

Jawaab - Single agent simpler hota hai; multi-agent tab jab specialization/handoffs clear value den.
Level: Basic

Sawal - Multi-agent failure?

Jawaab - Coordination overhead, duplicated work, inconsistent state, loops aur cost explosion.
Level: Normal

Sawal - Memory types?

Jawaab - Working, conversational, semantic, episodic aur durable long-term memory use cases ke hisaab se.
Level: Normal

Sawal - Agent ko arbitrary database access doge?

Jawaab - Nahi. Narrow, permission-aware tools use karunga.
Level: Normal

Sawal - Irreversible action ka rule?

Jawaab - Preview + validation + approval + idempotency + audit, especially high-impact actions.
Level: Normal

Sawal - Agent eval kaise karoge?

Jawaab - Goal success, tool correctness, policy compliance, safety, latency, cost aur failure recovery cases measure karunga.
Level: Normal

Sawal - Autonomous coding agent ka risk?

Jawaab - Code execution, credentials, network access aur destructive actions high-risk ho sakte hain; sandbox and approval needed.
Level: Advanced

Sawal - AVO kya idea hai?

Jawaab - Uploaded paper me agent variation operator ban kar previous solutions, knowledge, tools aur evaluation feedback se

autonomous iterative optimization karta hai.
Level: Advanced

MCP

Sawal - MCP kya hai?

Jawaab - AI applications ko tools, resources aur prompts ke through external systems se connect karne wala protocol.
Level: Basic

Sawal - MCP server kya expose kar sakta hai?

Jawaab - Tools, resources aur prompts jaise capabilities.
Level: Basic

Sawal - MCP client kya karta hai?

Jawaab - Host/agent ki taraf se server se capabilities discover aur invoke karta hai.
Level: Basic

Sawal - 2026-07-28 MCP spec me major change?

Jawaab - Stateless protocol core, header-based routing, cacheable list results aur authorization hardening.
Level: Basic

Sawal - MCP tool schema kyun?

Jawaab - Model ko valid arguments generate karne me help aur server-side validation possible hoti hai.
Level: Basic

Sawal - MCP security me kya important?

Jawaab - Explicit authorization, user consent for sensitive actions, least privilege, output validation, logging and trust boundaries.
Level: Normal

Sawal - MCP Tasks kya hai?

Jawaab - Asynchronous work ke liye task handle lifecycle, jisse client later result retrieve kar sakta hai.
Level: Normal

Sawal - MCP skills kya hain?

Jawaab - Current Skills extension resources ke through agent skills serve karne ka mechanism define karta hai; content ko untrusted prompt-injection surface treat karna chahiye.
Level: Normal

Data Science + Analysis

Sawal - EDA kya hai?

Jawaab - Dataset ko inspect, summarize aur visualize karke patterns, quality issues aur hypotheses identify karna.
Level: Basic

Sawal - Mean vs median?

Jawaab - Mean average hai; median ordered data ka middle point. Outliers ke against median more robust hota hai.
Level: Basic

Sawal - IQR kya hai?

Jawaab - Q3-Q1; middle 50% spread ko measure karta hai.
Level: Basic

Sawal - Confidence interval kya batata hai?

Jawaab - Sample-based estimate ki uncertainty range; exact population probability statement nahi.
Level: Basic

Sawal - p-value kya hai?

Jawaab - Null hypothesis ke under observed ya more extreme evidence ki probability measure.
Level: Basic

Sawal - Type I vs Type II error?

Jawaab - Type I = false positive; Type II = false negative.
Level: Normal

Sawal - Correlation vs causation?

Jawaab - Correlation association hai; causation ke liye stronger design/evidence chahiye.
Level: Normal

Sawal - Cross-validation kyun?

Jawaab - Model generalization estimate aur model selection me help karta hai.
Level: Normal

Sawal - Precision vs recall?

Jawaab - Precision predicted positives me kitne correct; recall actual positives me kitne capture kiye.
Level: Normal

Sawal - PR-AUC kab useful?

Jawaab - Rare positive class me precision-recall tradeoff ko summarize karne ke liye often useful.
Level: Normal

Sawal - Feature leakage kya hai?

Jawaab - Training feature me future/target-derived information aa jana, jis se offline score falsely high ho sakta hai.
Level: Advanced

Sawal - A/B test design me kya dekhoge?

Jawaab - Randomization, primary metric, sample size, guardrails, duration, interference aur novelty effects.
Level: Advanced

Data Engineering

Sawal - ETL vs ELT?

Jawaab - ETL me transform load se pehle; ELT me raw data warehouse/lake me load karke transform downstream.
Level: Basic

Sawal - Data lake vs warehouse?

Jawaab - Lake raw/flexible large-scale storage; warehouse curated analytical serving ke liye optimized.
Level: Basic

Sawal - Lakehouse kya hai?

Jawaab - Lake storage ke flexibility ko transactional table/analytics features ke saath combine karne ka architecture.
Level: Basic

Sawal - Fact vs dimension?

Jawaab - Fact measurable events; dimension descriptive entities/context.
Level: Basic

Sawal - CDC kya hai?

Jawaab - Source database changes ko capture karke downstream propagate karna.
Level: Basic

Sawal - Idempotency kya hai?

Jawaab - Same event/request retry hone par duplicate business effect na ho.

Level: Normal

Sawal - Kafka partition kya deta hai?

Jawaab - Parallelism aur per-partition ordering.
Level: Normal

Sawal - Consumer group kya hai?

Jawaab - Consumers ka coordinated set jahan partitions group members me distribute hote hain.
Level: Normal

Sawal - Event time vs processing time?

Jawaab - Event time business event ka timestamp; processing time pipeline ne kab handle kiya.
Level: Normal

Sawal - Watermark kya hai?

Jawaab - Late data/state ko bound karne ka mechanism in event-time streaming.
Level: Normal

Sawal - Data contract kya hai?

Jawaab - Schema, semantics, ownership, freshness/SLA aur change policy ka explicit agreement.
Level: Advanced

Sawal - Data quality ke core dimensions?

Jawaab - Completeness, validity, uniqueness, consistency, freshness, integrity.
Level: Advanced

Sawal - Airflow DAG kya hai?

Jawaab - Tasks aur dependencies ko workflow ke form me describe karta hai.
Level: Advanced

Sawal - Shuffle kya hota hai?

Jawaab - Distributed processing me data ko network across redistribute karna; expensive ho sakta hai.
Level: Advanced

Sawal - Data skew kya hai?

Jawaab - Kuch keys/partitions par disproportionate data load, jis se one task straggler ban sakta hai.
Level: Advanced

Production + Security

Sawal - Model monitoring me kya track karoge?

Jawaab - Quality, latency, error rate, drift, cost, throughput, safety signals.
Level: Basic

Sawal - Circuit breaker kya hai?

Jawaab - Repeated downstream failures me calls temporarily stop karke cascading failure prevent karta hai.
Level: Basic

Sawal - Rate limiting kyun?

Jawaab - Abuse, cost spikes aur overload control ke liye.
Level: Basic

Sawal - Canary deployment kya hai?

Jawaab - Small traffic slice ko new version par bhejna aur metrics compare karna.
Level: Basic

Sawal - Rollback kaise?

Jawaab - Versioned model/prompt/config artifacts aur reproducible deployment rakhna.
Level: Basic

Sawal - Secrets kahan?

Jawaab - Secret manager/environment injection; source code ya prompt me hard-code nahi.
Level: Normal

Sawal - Tenant isolation kya hai?

Jawaab - Har customer ka data/tool scope logically and technically separate rakhna.
Level: Normal

Sawal - NIST AI RMF kya karta hai?

Jawaab - AI trustworthiness/risk management ko design, development, use aur evaluation me incorporate karne ke liye framework deta hai.
Level: Normal

System Design

Sawal - Scalable AI service ka basic architecture?

Jawaab - API gateway -> auth/rate limit -> orchestrator -> model/retrieval/tools -> validation -> response, with cache/queue/observability.
Level: Basic

Sawal - Latency reduce kaise?

Jawaab - Streaming, caching, smaller model route, parallel tools, retrieval optimization aur fewer tokens.
Level: Basic

Sawal - Cost reduce kaise?

Jawaab - Model routing, caching, context pruning, batching, quotas aur per-workflow budgets.
Level: Basic

Sawal - High availability ke liye?

Jawaab - Stateless API replicas, managed durable stores, retries, timeouts, circuit breakers, health checks.
Level: Basic

Sawal - Async workload kab?

Jawaab - Long-running research, batch analysis, large document processing ya background agent tasks.
Level: Basic

Sawal - Queue kyun?

Jawaab - Spike absorb, retries, decouple producers/consumers aur backpressure ke liye.
Level: Normal

Sawal - Cache kya karoge?

Jawaab - Stable retrieval results, embeddings, metadata, repeated prompts where semantics remain valid.
Level: Normal

Sawal - Human approval kahan?

Jawaab - Refund, deletion, financial action, production changes, privileged tool calls jaise sensitive transitions par.
Level: Normal

Sawal - SLO kya hai?

Jawaab - Service level objective, e.g. latency/error/availability target jiske against reliability evaluate karte hain.
Level: Normal

Advanced Cross-Role Questions

Sawal - Model routing ko kaise design karoge?

Jawaab - Task complexity, latency budget, privacy, modality, context requirement, expected quality aur cost ko routing signals banaunga. Route decision deterministic rules + benchmark data se tune karunga.

Level: Advanced

Sawal - RAG me query rewriting kya hai?

Jawaab - User query ko retrieval-friendly form me transform karna, jaise missing keywords add karna ya ambiguous references clarify karna. Rewrite ko answer claim nahi samajhna chahiye.

Level: Advanced

Sawal - HyDE style retrieval ka idea kya hai?

Jawaab - Model se hypothetical relevant passage banwa kar uska embedding retrieve ke liye use karna. Isse semantic recall improve ho sakta hai, lekin hallucinated query drift ko evaluate karna zaroori hai.

Level: Advanced

Sawal - Parent-child retrieval kya solve karta hai?

Jawaab - Small child chunks precision improve karte hain aur parent document broader context provide karta hai. Retrieval unit aur generation context ko alag rakhne ka pattern hai.

Level: Advanced

Sawal - RAG me ACL kaise preserve karoge?

Jawaab - Ingestion par permission metadata attach karo, retrieval me mandatory authorization filters lagao, aur tool/data access layer par second enforcement rakho.

Level: Security

Sawal - RAG freshness SLA kya hota hai?

Jawaab - Define karo ki source update ke kitne time andar index reflect karega. Freshness timestamp monitor karo aur stale results par fallback ya warning design karo.

Level: Production

Sawal - Retriever recall high but answer quality low kyun?

Jawaab - Relevant document retrieved ho sakta hai lekin context packing, ordering, contradictory sources, model instruction ya evidence selection fail ho sakte hain.

Level: Advanced

Sawal - Context packing kya hai?

Jawaab - Retrieved evidence ko order, deduplicate, compress aur source metadata ke saath model context me arrange karna.

Level: Advanced

Sawal - Lost-in-the-middle problem ko kaise reduce karoge?

Jawaab - Relevant evidence ko carefully position karo, context size limit karo, summarize/rerank karo aur critical evidence ko final prompt structure me high-salience placement do.

Level: Advanced

Sawal - RAG me duplicate chunks problem?

Jawaab - Same semantic content ko near-duplicate removal, MMR/diversity selection aur document-level dedupe se reduce karo.

Level: Advanced

Sawal - Tool latency aggregation kaise samjho?

Jawaab - Sequential calls latency add karte hain; independent calls parallel ho sakte hain. Trace spans se critical path identify karo.

Level: Production

Sawal - Tool timeout ka default behavior?

Jawaab - Bounded timeout, cancellation, retry only for safe/idempotent operations, aur user-facing degraded response define karo.

Level: Production

Sawal - Retry storm kya hai?

Jawaab - Multiple layers ek hi failure ko retry karein to traffic multiply ho sakta hai. Central retry policy, jitter, budgets aur

circuit breakers use karo.
Level: Advanced

Sawal - Idempotency key ka practical example?

Jawaab - Payment/refund/order-write request me client-generated key store karo; same key repeat hone par same prior result return karo.
Level: Production

Sawal - Saga pattern AI workflows me kab?

Jawaab - Multi-step distributed side effects me transaction impossible ho to compensating actions ke saath saga-style workflow use kar sakte ho.
Level: Advanced

Sawal - Human approval ko model se kyun nahi karwana?

Jawaab - Approval ka purpose independent control boundary banana hai; same model ko apne action ko approve karne dena separation of duties weak karta hai.
Level: Security

Sawal - Agent budget kya hai?

Jawaab - Per-run limits such as max steps, tool calls, tokens, wall-clock time and spend. Budget exceed par controlled stop/fallback.
Level: Advanced

Sawal - Agent termination conditions?

Jawaab - Goal satisfied, verifier pass, explicit user stop, max budget, deadline, repeated-state/cycle detection ya safety policy block.
Level: Advanced

Sawal - Verifier agent ka role?

Jawaab - Generated result ya action ko independent criteria ke against check karna. Verifier ko same failure mode reproduce karne se bachane ke liye diverse checks useful hain.
Level: Advanced

Sawal - Critic vs verifier difference?

Jawaab - Critic suggestions/quality feedback deta hai; verifier binary/structured validity decision de sakta hai. Real system me dono combine ho sakte hain.
Level: Advanced

Sawal - Agent memory me privacy issue?

Jawaab - Har memory item ko user consent, retention, purpose, access scope aur delete/update semantics ke through govern karo.
Level: Security

Sawal - Memory poisoning kya hai?

Jawaab - Malicious ya incorrect stored memory future decisions influence kare. Write validation, provenance, confidence and user-visible memory controls useful hain.
Level: Security

Sawal - Multi-agent communication protocol kya define kare?

Jawaab - Message schema, sender identity, allowed recipients, payload limits, correlation ID, timeout, retry policy and termination semantics.
Level: Advanced

Sawal - Supervisor agent bottleneck kaise reduce?

Jawaab - Parallel specialization, static routing for obvious cases, bounded delegation and batching. Supervisor ko every micro-step decide na karne do.

Level: Advanced

Sawal - Agent observability me trace hierarchy?

Jawaab - Request trace ke andar model, retrieval, tool, policy, verifier aur external API spans. Har span me latency, status aur safe metadata.
Level: Production

Sawal - Quality regression after model upgrade?

Jawaab - Golden eval suite rerun, segmented slices compare, critical business flows gate karo aur canary deployment use karo.
Level: LLMOps

Sawal - Prompt regression after tool schema change?

Jawaab - Tool-call eval cases aur contract tests rerun karo; invalid-argument rate, tool selection accuracy and downstream outcomes compare karo.
Level: LLMOps

Sawal - Eval set leakage kya hai?

Jawaab - Agar test cases prompt/training/optimization loop me repeatedly expose ho jayein to test score unrealistic ho sakta hai. Holdout sets maintain karo.
Level: Advanced

Sawal - Synthetic eval data ka risk?

Jawaab - Synthetic data coverage fast badhata hai, lekin distribution bias ya unrealistic cases aa sakte hain. Human-labeled holdout mix karo.
Level: Advanced

Sawal - LLM-as-a-judge ka limitation?

Jawaab - Judge bhi model bias/error carry kar sakta hai. Rubric, calibration, pairwise checks and human sampling se validate karo.
Level: Advanced

Sawal - Data scientist ke liye calibration kyun?

Jawaab - Predicted probabilities ko observed frequencies ke saath align karne ke liye. Risk thresholds aur decision policies me calibration useful hai.
Level: Data Science

Sawal - Precision-recall threshold kaise choose?

Jawaab - Business false-positive vs false-negative cost ko define karke validation set par threshold tune karo.
Level: Data Science

Sawal - Drift alert kaise avoid false positives?

Jawaab - Seasonality, expected promotions aur known distribution shifts model karo; baseline window aur alert hysteresis use karo.
Level: LLMOps

Sawal - Feature store ka core idea?

Jawaab - Reusable, governed features ko consistent training aur online inference use cases ke liye manage karna. Offline/online consistency important hai.
Level: Data Engineering

Sawal - Point-in-time correctness kya hai?

Jawaab - Historical training row ke timestamp par jo information available thi sirf wahi join honi chahiye; future leakage avoid karta hai.
Level: Data Engineering

Sawal - Backfill kaise safe karoge?

Jawaab - Idempotent transforms, versioned inputs, partition-scoped reruns, data quality checks and downstream impact review.
Level: Data Engineering

Sawal - Late-arriving dimension kya problem hai?

Jawaab - Fact event dimension record se pehle aa sakta hai; pipeline ko surrogate/default key ya delayed enrichment strategy chahiye.
Level: Data Engineering

Sawal - Slowly changing dimension Type 2?

Jawaab - Har historical change ka new row/version maintain karke valid_from/valid_to jaise timestamps se history preserve karna.
Level: Data Engineering

Sawal - Partition pruning kya hai?

Jawaab - Query filters se irrelevant partitions/files skip karna, scan cost reduce karta hai.
Level: Data Engineering

Sawal - Small files problem kya hai?

Jawaab - Bahut chhoti files metadata overhead aur read inefficiency create karti hain. Compaction/optimize strategy use hoti hai.
Level: Data Engineering

Sawal - Data skew ka fix?

Jawaab - Salting, skew-aware joins, repartitioning, broadcast where appropriate, and key redesign depending on workload.
Level: Data Engineering

Sawal - CDC delete handling?

Jawaab - Delete events preserve karo ya tombstone semantics use karo; sink table ko delete/soft-delete policy ke saath synchronize karo.
Level: Data Engineering

Sawal - Schema registry kyun?

Jawaab - Event schemas ko version aur compatibility rules ke saath manage karne ke liye.
Level: Data Engineering

Sawal - Backward compatible schema change?

Jawaab - Naya producer/consumer old clients ko break na kare. Optional fields and compatible type evolution preferred.
Level: Data Engineering

Sawal - Data observability kya track kare?

Jawaab - Freshness, volume, distribution, schema, nulls, duplicates, lineage and downstream incidents.
Level: Data Engineering

Sawal - Semantic layer kyun?

Jawaab - Business metrics ko consistent definitions aur reusable dimensions ke saath expose karne ke liye, so BI and AI agree on meaning.
Level: Data Analytics

Sawal - Metric contract kya hai?

Jawaab - Metric name, formula, grain, filters, owner, update cadence and source-of-truth explicitly define karna.
Level: Data Analytics

Sawal - Funnel drop-off kaise diagnose?

Jawaab - Step-wise conversion, segment breakdown, cohort comparison and event instrumentation audit. Aggregate rate se cause assume mat karo.

Level: Data Analytics

Sawal - Cohort analysis kya?

Jawaab - Users ko common start event/time ke basis par group karke retention/revenue behavior over time compare karna.
Level: Data Analytics

Sawal - RFM analysis kya?

Jawaab - Recency, frequency, monetary value ke basis par customers segment karna.
Level: Data Analytics

Sawal - A/B test interference kya hai?

Jawaab - One user ka treatment doosre users ke outcome ko affect kare. Network effects/social features me independent-unit assumption fail ho sakti hai.
Level: Data Science

Sawal - Experiment guardrail metric?

Jawaab - Primary goal ke saath safety/business health metric, jaise churn, latency, refunds or support tickets, jisse harmful trade-off detect ho.
Level: Data Science

Sawal - Causal inference vs prediction?

Jawaab - Prediction future outcome estimate karta hai; causal inference treatment/intervention effect estimate karta hai. Objective aur assumptions different hain.
Level: Data Science

Sawal - Feature importance vs causality?

Jawaab - Important predictive feature hona causal effect prove nahi karta. Correlated proxy bhi high importance de sakta hai.
Level: Data Science

Sawal - Data engineer aur AI engineer handoff?

Jawaab - Data engineer reliable contracts, quality, lineage and serving interfaces deta hai; AI engineer retrieval/model/tool logic ko consume karke application behavior build karta hai.
Level: Cross-role

Sawal - Production schema migration ka safe approach?

Jawaab - Versioned migration, backward compatibility window, dual-read/write where needed, validation and rollback plan.
Level: System Design

Sawal - Canary AI deployment me kya compare?

Jawaab - Error rate, task success, safety violations, latency, cost, tool-call failures and business KPIs by traffic slice.
Level: Production

Sawal - Disaster recovery for AI system?

Jawaab - Versioned artifacts, durable data, backups, restore drills, fallback provider/model, RPO/RTO definitions and tested runbooks.
Level: Production

Sawal - RPO vs RTO?

Jawaab - RPO = kitna data loss acceptable; RTO = service ko kitni der me restore karna target hai.
Level: System Design

Sawal - Blue-green vs canary?

Jawaab - Blue-green full environment switch pattern; canary gradual traffic exposure pattern. Risk and rollback workflow differ.
Level: System Design

Sawal - Load shedding kya hai?

Jawaab - Overload me low-priority work reject/defer karke critical traffic preserve karna.
Level: System Design

Sawal - Backpressure kya hai?

Jawaab - Fast producer ko slow consumer ke rate ke according control karna; queues, flow control and admission limits use hote hain.
Level: Data Engineering

Sawal - Queue ordering guarantee kaise choose?

Jawaab - Business requirement ke hisaab se global ordering, partition/key ordering or no ordering. Strong ordering throughput reduce kar sakta hai.
Level: Data Engineering

Sawal - Exactly-once ko interview me kaise explain?

Jawaab - End-to-end property hai, sirf broker feature nahi. Usually idempotent processing, transactional writes and replay-safe design se practical exactly-once effects achieve karte hain.
Level: Advanced

Sawal - SQL window function kab?

Jawaab - Running totals, ranking, lag/lead, retention calculations aur row-level analytics ke liye.
Level: Data Analytics

Sawal - AI analytics copilot me SQL safety?

Jawaab - Read-only role, allowlisted schemas, query parser/validator, row/scan limits, timeout and result-size caps.
Level: Security

Sawal - Text-to-SQL hallucination ka control?

Jawaab - Semantic schema context + constrained generation + SQL parser + allowlist + dry-run/EXPLAIN + result validation.
Level: AI + Data

Sawal - Agentic Text-to-SQL ka extra risk?

Jawaab - Agent query generate, execute aur iterate kar sakta hai; execution permissions aur budget ko tightly gate karo.
Level: Security

Sawal - Why not expose raw DB credentials to agent?

Jawaab - Credentials broad privilege aur data exfiltration risk create karte hain. Scoped service identity/tool gateway safer boundary hai.
Level: Security

Sawal - Tool gateway kya hai?

Jawaab - Central policy-enforcing layer jo tool definitions, auth, rate limits, logging, schema validation aur routing manage karti hai.
Level: Architecture

Sawal - Policy engine ka role?

Jawaab - Allowed actions ko user, tenant, environment, tool and risk level ke basis par decide karna.
Level: Security

Sawal - Auditability ka minimum?

Jawaab - Who, what, when, which policy, which tool, outcome and approval - safe metadata ke saath traceable record.
Level: Governance

Sawal - Reproducibility kya?

Jawaab - Same model/prompt/tool/data/config versions ke saath result behavior ko rerun aur compare kar pana.
Level: LLMOps

Sawal - Artifact registry me kya store?

Jawaab - Model/version, prompt/config, evaluation results, dependency metadata, deployment status and rollback target.
Level: MLOps

Sawal - Embedding index reproducibility?

Jawaab - Embedding model/version, chunker config, preprocessing, metadata schema and index parameters version karo.
Level: RAG

Sawal - Agent benchmark kaise banega?

Jawaab - Representative task set + expected outcomes + allowed tools + success criteria + adversarial cases + cost/latency budget.
Level: Advanced

Sawal - Tool selection accuracy kaise measure?

Jawaab - Expected tool vs selected tool compare karo, wrong-tool rate aur downstream success rate separately track karo.
Level: Advanced

Sawal - Agent handoff loop kaise detect?

Jawaab - Correlation IDs and state history se repeated agent transitions detect karo; max handoffs and cycle rules lagao.
Level: Advanced

Sawal - MCP tool catalog cache kyun?

Jawaab - Tool discovery response cache karne se repeated listing overhead reduce ho sakta hai; current MCP 2026 spec list responses ko cache hints ke saath support karta hai.
Level: MCP

Sawal - MCP stateless architecture ka benefit?

Jawaab - Ordinary HTTP load balancing aur horizontal scaling easier ho sakta hai because protocol session affinity requirement remove hui.
Level: MCP

Sawal - MCP skill content ko trusted instructions maana chahiye?

Jawaab - Nahi. Current Skills extension explicitly prompt-injection risk ke perspective se served skill content ko untrusted input treat karti hai.
Level: MCP Security

Sawal - MCP server trust model?

Jawaab - Connected server ko automatically trusted business authority mat samjho; origin, authorization, capability scope and user approvals define karo.
Level: MCP Security

Sawal - Long-running agent ko kaise resume?

Jawaab - Checkpointed state + deterministic resume boundary + idempotent side effects + run identity use karo.
Level: Agentic

Sawal - Background agent UX kya hona chahiye?

Jawaab - Immediate acknowledgement + job ID/status + progress events + cancel/retry + final result notification.
Level: Product

Sawal - AI system ka error budget?

Jawaab - Reliability budget define karo; frequent errors/retries/cost spikes par feature rollout slow ya rollback kar sakte ho.
Level: Production

Sawal - Agent cost attribution kaise?

Jawaab - Trace level par model tokens, tool/API cost, retrieval calls and compute time aggregate karke tenant/project attribution karo.

Level: LLMOps

Sawal - Evaluation me segmentation kyun?

Jawaab - Average score hidden failures hide kar sakta hai. Language, customer tier, task type, document type and adversarial slice by slice inspect karo.

Level: Evaluation

Sawal - Human review sampling kaise?

Jawaab - Random + risk-based sampling combine karo. High-impact/low-confidence outputs ko higher review probability do.

Level: Evaluation

Sawal - What is a safe fallback?

Jawaab - Aisa deterministic/degraded response jo wrong action na le aur user ko next safe step de. Fallback ko evaluate aur monitor karo.

Level: Reliability

19 PROJECT DEFENSE + BEHAVIORAL + RESUME STORIES

Project defend karne ka formula: Problem -> Architecture -> Why this stack -> Data flow -> Failure -> Metrics -> Security -> Trade-off -> Demo.

90-second project answer template

```
Problem: kis business pain ko solve kiya?
Users: kaun use karega?
Architecture: input -> processing -> model/RAG/agent -> output
My work: maine personally kya build kiya?
Tech: why these components?
Evaluation: kaise prove kiya?
Failure: kya toot sakta hai?
Security: kya controls hain?
Next: kya improve karunga?
```

Project defense questions

Sawal - Aapne LangChain kyun use kiya?

Jawaab - Tool/model integration aur agent composition fast karne ke liye. Jahan complex stateful orchestration required hai wahan LangGraph add kiya.

Level: Defense

Sawal - RAG kyun, fine-tuning kyun nahi?

Jawaab - Knowledge frequently change hoti hai to retrieval better fit hai; fine-tuning behavior adaptation ke liye consider kiya.

Level: Defense

Sawal - Aapka agent galat tool call kare to?

Jawaab - Schema validation, authorization, retry/repair, verifier aur approval gate use karta hoon.

Level: Defense

Sawal - Latency high aaye to?

Jawaab - Trace se bottleneck identify karunga; retrieval/model/tool calls ko parallelize/cache/route karunga.

Level: Defense

Sawal - Production me failover?

Jawaab - Timeouts, retries, fallback model/provider, degraded response and queue-based recovery design karunga.

Level: Defense

Sawal - Aap quality kaise measure karte ho?

Jawaab - Golden eval set, task correctness, groundedness, tool correctness, safety, latency aur cost metrics use karta hoon.

Behavioral STAR skeleton

30. Situation: context 1-2 lines.
31. Task: exact responsibility.
32. Action: tumne kya kiya; tools and decisions.
33. Result: measurable/observable impact.
34. Learning: next time kya improve karoge.

20 CHECKPOINT + REVISION PLAN + MASTER CHECKLIST

Is section ko weekly mastery test ki tarah use karo.

50-question checkpoint

Sawal - Checkpoint 1: LLM, tokenization, embedding, attention aur context window explain karo.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 2: RAG me chunking, embedding, retrieval aur reranking ka role batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 3: Hybrid retrieval kab use karoge?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 4: Groundedness aur answer correctness me difference batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 5: Prompt injection kya hai aur controls kya hain?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 6: Structured output ka fayda batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 7: LangChain `create_agent` ka concept samjhao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 8: Middleware ka role batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 9: LangGraph ko LangChain se kaise differentiate karoge?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 10: State, persistence aur HITL explain karo.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 11: Agent loop aur workflow me difference batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 12: Multi-agent kab use karna hai?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 13: Least-privilege tool design ka example do.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.

Level: Checkpoint

Sawal - Checkpoint 14: Agent infinite loop ko kaise rokoge?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 15: MCP ka host-client-server model batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 16: MCP tools aur resources me difference batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 17: 2026 MCP stateless change kyun important hai?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 18: MCP security controls batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 19: Data analyst aur data scientist role difference batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 20: Mean/median/IQR explain karo.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 21: Confidence interval kya hai?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 22: Type I vs Type II error?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 23: Precision vs recall?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 24: Feature leakage?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 25: ETL vs ELT?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 26: Warehouse vs lake?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 27: Lakehouse?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 28: Fact vs dimension?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 29: CDC?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 30: Idempotency?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 31: Kafka partition?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 32: Consumer group?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 33: Event time vs processing time?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 34: Watermark?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 35: Shuffle?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 36: Data skew?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 37: Data contract?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 38: Data quality ke dimensions?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 39: Airflow DAG/task/scheduler?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 40: MLOps lifecycle?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 41: LLMOps metrics?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 42: Tracing me kya capture karoge?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 43: Model drift vs data drift?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 44: Canary deployment?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 45: Circuit breaker?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 46: Rate limiting?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 47: Cost optimization ideas?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 48: AI system ko secure kaise karoge?

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 49: Project ko 90 sec me explain karo.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 50: Apne project ka top failure mode aur fix batao.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Sawal - Checkpoint 51: Apne project ka ek measurable evaluation metric define karo.

Jawaab - Answer verbally first. Phir is master PDF ke related section se verify karo.
Level: Checkpoint

Weekly mastery cycle

35. Day 1: Learn one system topic + notes.
36. Day 2: Coding lab + edge cases.
37. Day 3: 10 interview answers aloud.
38. Day 4: One system-design diagram from memory.
39. Day 5: Project implementation.
40. Day 6: Debug + evaluation + security test.
41. Day 7: Checkpoint + weak-topic revision.

Master checklist

☐ LLM fundamentals
☐ Transformer/attention
☐ Prompt/context engineering
☐ Structured output/tool schemas
☐ Basic/advanced RAG

☐ LangChain
☐ LangGraph
☐ Agents + memory
☐ MCP
☐ Multi-agent systems
☐ Evaluation
☐ LangSmith/observability
☐ AI security
☐ System design
☐ Data analysis
☐ Statistics
☐ ML fundamentals
☐ SQL analytics
☐ Data modeling
☐ Kafka
☐ Spark
☐ Airflow concepts
☐ Lakehouse
☐ Data quality
☐ MLOps/LLMOps
☐ Docker/Kubernetes basics
☐ 3 end-to-end projects
☐ Project defense
☐ Mock interview

21 DATA ANALYST KA ADVANCED TOOLKIT

Business question ko data se measurable decision tak le jana seekho: SQL + KPI + funnel + cohort + dashboard + storytelling.

Data Analyst ka complete workflow

42. Business question ko exact metric me translate karo.
43. Source tables, grain aur join keys verify karo.
44. Data quality checks chalao.
45. SQL se analysis nikalo.
46. Segment/cohort/funnel se pattern isolate karo.
47. Visualization se signal clear karo.
48. Business impact aur next action explain karo.
49. Metric definitions document karo.

SQL me must-know topics

- INNER/LEFT/RIGHT/FULL joins aur duplicate-row diagnosis
- CTE aur nested query
- Window functions: ROW_NUMBER, RANK, LAG, LEAD, SUM OVER
- Conditional aggregation
- Date/time analytics
- Funnel conversion
- Cohort retention
- Deduplication
- Top-N per group
- Rolling averages
- Percentiles
- NULL handling
- Query plan aur indexing basics
- Read-only analytics roles aur query limits

SQL example - Top product per region

```
WITH ranked AS (
 SELECT
  region, product_id, revenue,
  ROW_NUMBER() OVER (
   PARTITION BY region
   ORDER BY revenue DESC
  ) AS rn
 FROM monthly_sales
)
SELECT region, product_id, revenue
FROM ranked
WHERE rn = 1;
```

Samjho: pehle region ke andar revenue ke basis par ranking hui; phir rn = 1 se har region ka top product mila.

SQL example - 7 day rolling average

```
SELECT event_date, revenue,
    AVG(revenue) OVER (
     ORDER BY event_date
     ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
    ) AS rolling_7d
FROM daily_revenue
ORDER BY event_date;
```

KPI aur metric design

- North-star metric aur supporting metrics alag rakho.

- Metric ka grain define karo: user, order, session, day etc.
- Denominator explicitly define karo.
- Window, filter, timezone aur late-data behavior document karo.
- Business owner aur source-of-truth table define karo.
- AI analytics systems me semantic layer use karke same metric definitions reuse karo.

Dashboard design

- Overview -> trend -> segment -> drill-down flow rakho.
- 20 charts ka dump nahi; decision-oriented view banao.
- Anomaly highlight aur alert context do.
- Raw numbers ke saath interpretation aur action bhi dikhao.
- Power BI/Tableau jaise tools me calculated measures, filters, parameters aur drill-through ka use samjho.

Data Analyst interview Q&A

Sawal - SQL me LEFT JOIN ke baad row count achanak kyun badh sakta hai?

Jawaab - Right table me one-to-many match ho sakta hai. Grain check karo aur join key uniqueness validate karo.
Level: Basic

Sawal - WHERE aur HAVING me difference?

Jawaab - WHERE rows ko grouping se pehle filter karta hai; HAVING aggregate/group result ko filter karta hai.
Level: Basic

Sawal - ROW_NUMBER aur RANK me difference?

Jawaab - ROW_NUMBER unique sequence deta hai; RANK ties par same rank de sakta hai aur gaps aa sakte hain.
Level: Normal

Sawal - Cohort analysis kya solve karta hai?

Jawaab - Same start event/time wale users ko group karke retention, revenue ya engagement ko time ke saath compare karta hai.
Level: Normal

Sawal - Funnel analysis me sabse common mistake kya hai?

Jawaab - Different events ko inconsistent user/session grain par compare karna. Har stage ka definition aur denominator clear rakho.
Level: Normal

Sawal - Dashboard me metric conflict kaise solve karoge?

Jawaab - Single semantic definition, owner, source-of-truth, refresh rule aur metric contract publish karo.
Level: Advanced

Sawal - Data me outlier dekh kar remove kar doge?

Jawaab - Automatically nahi. Pehle identify karo ki outlier error hai, rare valid event hai ya segment behavior.
Level: Normal

Sawal - A/B test dashboard me kya dikhaoge?

Jawaab - Primary metric, confidence interval, guardrails, sample size, segment breakdown aur experiment status.
Level: Advanced

Sawal - SQL query slow hai to pehla step kya?

Jawaab - Query plan dekho, scan volume/join order/index/partition pruning aur unnecessary columns check karo.
Level: Advanced

Sawal - AI analytics copilot ko raw DB access doge?

Jawaab - Nahi. Read-only scoped identity, allowlisted schemas, parser/validator, row/scan limits aur timeout use karunga.

Level: Advanced

22 DATA SCIENTIST KA ADVANCED TOOLKIT

Statistics se leke experiments, ML modeling, calibration, time series, NLP aur business impact tak complete thinking build karo.

DS problem framing

50. Outcome aur decision define karo.
51. Prediction vs explanation vs causal question separate karo.
52. Target leakage aur label availability check karo.
53. Success metric ko business cost se connect karo.
54. Offline evaluation aur online validation design karo.

Statistics - interview depth

- Mean, median, variance, standard deviation, percentile, quartile, IQR
- Probability rules aur conditional probability
- Distributions: normal, binomial aur use-case based choices
- Sampling, bias, confounding
- Confidence interval
- Hypothesis test, p-value, significance level
- Type I/II errors aur power
- Two-sample/paired t-test
- Chi-square independence/goodness-of-fit
- Regression aur nonlinear relationships
- Effect size vs statistical significance

Hypothesis test ka practical flow

55. H0 aur H1 define karo.
56. Metric aur unit of analysis define karo.
57. Sample-size/power thinking karo.
58. Test statistic calculate karo.
59. p-value + confidence interval interpret karo.
60. Effect size aur business meaning explain karo.
61. Multiple testing/peeking risks consider karo.

ML evaluation

- Classification: precision, recall, F1, ROC-AUC, PR-AUC, calibration
- Regression: MAE, RMSE, MAPE/SMAPE where appropriate
- Ranking/recommendation: Precision@K, Recall@K, NDCG
- Imbalanced data me threshold selection
- Calibration aur probability quality
- Slice-wise evaluation: language, region, segment, task type

Data leakage ka practical example

```
TRAINING SNAPSHOT: 2025-01-31
feature = customer_total_spend_as_of_2025_02-28

PROBLEM:
Feature me future information aa gayi.

SAFE VERSION:
customer_total_spend_as_of_event_time
```

Point-in-time feature construction ka goal ye hai ki training row me wahi information aaye jo prediction ke timestamp par genuinely available thi.

Time series ke must-know ideas

- Trend, seasonality, cyclical behavior
- Temporal split - random split nahi
- Lag features and rolling windows
- Forecast horizon
- Backtesting
- Data leakage from future periods
- Exogenous variables
- Drift and recalibration

Recommendation system basics

- Candidate generation -> ranking -> re-ranking
- Collaborative vs content-based signals
- Cold start
- Popularity bias
- Offline vs online metrics
- Diversity/novelty/coverage trade-offs

NLP + GenAI bridge

- Text classification/sentiment -> embeddings -> retrieval -> generation
- Entity extraction and document tagging
- Chunking + metadata for enterprise corpora
- Human review for high-risk outputs

Data Scientist interview Q&A

Sawal - Prediction aur causal inference me difference?

Jawaab - Prediction future outcome ko accurately estimate karta hai; causal inference intervention ka effect estimate karta hai. Assumptions aur design dono different hote hain.

Level: Basic

Sawal - Cross-validation kab use nahi karna chahiye?

Jawaab - Time series ya grouped/related observations me random folds leakage create kar sakte hain; structure-aware split chahiye.

Level: Advanced

Sawal - PR-AUC imbalanced data me useful kyun?

Jawaab - Rare positive class par precision-recall trade-off ko focus karta hai; accuracy aur ROC-AUC ke saath context me interpret karo.

Level: Normal

Sawal - Calibration kya hai?

Jawaab - Predicted probabilities aur actual event frequencies ke alignment ko measure karna.

Level: Advanced

Sawal - Feature importance causality prove karta hai?

Jawaab - Nahi. Feature predictive signal ho sakta hai ya proxy/confounder ho sakta hai.

Level: Normal

Sawal - A/B test me sample ratio mismatch kya signal ho sakta hai?

Jawaab - Randomization ya traffic assignment problem ka signal ho sakta hai; experiment instrumentation audit karo.

Level: Advanced

Sawal - p-value < 0.05 ka matlab kya?

Jawaab - Null hypothesis true hone ki probability simply 5% nahi hoti. It means H0 ke under observed ya more-extreme evidence ka tail probability threshold se chhota hai, given test setup.

Level: Advanced

Sawal - Model score high hai par production weak kyun?

Jawaab - Distribution shift, leakage, threshold mismatch, poor calibration, label delay, latency or business behavior change ho sakta hai.

Level: Advanced

Sawal - Class imbalance ka basic fix?

Jawaab - Metric choice, thresholding, class weights/resampling aur business cost matrix consider karo; blind oversampling nahi.

Level: Normal

Sawal - Time series forecast me random shuffle problem?

Jawaab - Future observations train me aa sakte hain, jisse validation falsely optimistic ho jata hai.

Level: Normal

23 DATA ENGINEER KA ADVANCED TOOLKIT

Reliable AI ke neeche reliable data platform hona chahiye: ingestion, CDC, warehouse/lakehouse, streaming, orchestration, quality, lineage aur serving.

End-to-end data platform

```
Sources -> CDC / Events / Files / APIs
      |
    Kafka / Object Storage
      |
   Bronze -> Silver -> Gold
      |
  Lakehouse / Warehouse
   |      |
  BI/DS   AI/RAG
   |      |
  Metrics  Retrieval
```

Batch engineering

- Partitioned ingestion
- Incremental processing
- Backfill strategy
- Late-arriving data
- Idempotent writes
- Retry-safe transformations
- Dependency-aware orchestration

Streaming engineering

- Producer -> topic -> partition -> consumer group
- Event time vs processing time
- Watermarks
- Late events
- Stateful processing
- Deduplication
- Backpressure
- Dead-letter queue
- Replay from offsets

CDC aur SCD

```
SOURCE CDC
+ INSERT
+ UPDATE
+ DELETE
   |
VALIDATE -> ORDER -> APPLY
   |
TARGET TABLE
   |
SCD2: valid_from / valid_to / is_current
```

Lakehouse engineering

- ACID tables on object storage
- Schema enforcement and evolution
- Time travel/versioning
- MERGE/upsert for CDC
- Partitioning and pruning

- Compaction / small-file control
- Retention and snapshot maintenance

Current Delta Lake docs ACID transactions, streaming + batch unification, schema enforcement, time travel aur merge/update/delete support describe karti hain. Apache Iceberg docs schema aur partition evolution ke saath snapshots/time travel ko support karti hain.

dbt / transformation layer

- Model = curated transformation
- Incremental models for large datasets
- Tests for business/data assumptions
- Documentation + lineage
- Source freshness
- Semantic definitions

Incremental model ka logic

```
FULL RUN:
all valid rows process karo

INCREMENTAL RUN:
watermark / updated_at se new-or-changed rows lo
+ unique_key se merge/upsert
+ periodic full refresh strategy
```

Orchestration - Airflow mental model

- DAG = workflow definition
- Task = unit of work
- Dependencies = execution order
- Scheduler = ready work ko schedule karta hai
- Retries/timeouts = reliability controls
- Backfill = historical interval ko safely reprocess karna
- Tasks ko deterministic aur retry-safe rakhna zaroori hai

Data quality framework

- Completeness
- Uniqueness
- Validity
- Consistency
- Freshness
- Referential integrity
- Distribution drift
- Schema compatibility
- Business-rule validation

SQL data-quality gate

```
SELECT
 COUNT(*) AS rows,
 COUNT(*) FILTER (WHERE customer_id IS NULL) AS null_ids,
 COUNT(*) - COUNT(DISTINCT order_id) AS duplicate_id_count
FROM orders;
```

Production gate ka rule: quality check fail ho to downstream publish ko automatically block/quarantine/alert pattern use karo. Great Expectations jaisi frameworks Expectations ko verifiable data assertions aur production validation workflow ke saath use karti hain.

Sawal - Data SLA aur SLO me difference kya?

Jawaab - SLA external commitment ho sakta hai; SLO internal reliability target. Example: data 99% days me 7 AM se pehle

fresh ho.
Level: Normal

Sawal - Lineage kyun important hai?

Jawaab - Column/table change se kaunse downstream dashboards, models ya AI indexes impact honge, ye quickly trace karne ke liye.
Level: Advanced

Sawal - Vector index ko data pipeline ka part kyun samjhenge?

Jawaab - RAG me corpus change hone par chunks, embeddings, metadata aur index refresh coordinated hona chahiye; otherwise stale retrieval ho sakta hai.
Level: Advanced

Sawal - Data reconciliation ka pattern?

Jawaab - Source vs target row counts, sums, key sets aur sampled records compare karo, especially migration/backfill ke baad.
Level: Advanced

Feature store aur training-serving skew

- Offline store: historical training data
- Online store: low-latency serving
- Point-in-time joins
- Feature freshness
- Training-serving consistency
- Feature ownership/versioning

Data Engineer interview Q&A

Sawal - Idempotent pipeline kya hai?

Jawaab - Same input ko retry karne par final business state duplicate nahi banta. Keys, merge/upsert aur deterministic transformations help karte hain.
Level: Basic

Sawal - CDC me delete kaise handle karoge?

Jawaab - Tombstone/delete event capture karo aur downstream sink me hard-delete ya soft-delete policy apply karo.
Level: Normal

Sawal - Kafka partition ka main purpose?

Jawaab - Parallelism aur per-partition ordering provide karna. Global ordering expensive aur usually unnecessary hoti hai.
Level: Basic

Sawal - Consumer group kya karta hai?

Jawaab - Same topic ko multiple consumers me partitions distribute karta hai so processing parallel ho sake.
Level: Basic

Sawal - Watermark kyun?

Jawaab - Event-time streaming me late data ke against state growth ko bound karne aur completeness trade-off manage karne ke liye.
Level: Advanced

Sawal - Exactly-once ko kaise explain karoge?

Jawaab - End-to-end property samjho. Broker alone enough nahi; idempotent processing, transactional writes aur replay-safe design zaroori hai.
Level: Advanced

Sawal - Data skew ka impact?

Jawaab - Kuch partitions huge ho kar straggler tasks create karte hain. Key redesign, salting, repartitioning ya selective broadcast help kar sakte hain.
Level: Advanced

Sawal - Small-files problem kya?

Jawaab - Too many tiny objects/files metadata aur scan overhead badhate hain; compaction/optimized write strategy help karti hai.
Level: Normal

Sawal - Backfill safe kaise?

Jawaab - Partition-scoped rerun, versioned inputs, idempotent transforms, isolated outputs aur downstream compatibility checks.
Level: Advanced

Sawal - SCD Type 2 kya?

Jawaab - Dimension ke historical versions ko separate rows me valid_from/valid_to ya equivalent validity columns se track karna.
Level: Normal

Sawal - Lakehouse kya solve karta hai?

Jawaab - Object-storage scale/flexibility ke saath transactional tables, schema controls, time travel aur analytical serving patterns combine karta hai.
Level: Normal

Sawal - dbt incremental model kyun?

Jawaab - Large table ko har run full recompute karne ke bajay new/changed slice process karke runtime/cost reduce karna.
Level: Normal

Sawal - Data contract kya?

Jawaab - Schema + semantics + ownership + freshness/SLA + compatibility/change policy ka explicit agreement.
Level: Advanced

Sawal - Feature leakage kya?

Jawaab - Training feature me future information aa jana. Point-in-time correctness se avoid karna chahiye.
Level: Advanced

Sawal - Data observability kya dekhta hai?

Jawaab - Freshness, volume, schema, distributions, nulls, duplicates, lineage aur downstream incident signals.
Level: Advanced

24 AGENTIC AI KA ADVANCED MASTERCLASS

Agent ko chat wrapper mat samjho. Goal, state, tools, policy, memory, planning, execution, verification aur recovery ko ek system ke roop me design karo.

Agent ke 10 core building blocks

62. Goal / objective
63. State
64. Model
65. Tools
66. Memory
67. Planner
68. Executor
69. Verifier / critic
70. Policy / guardrails
71. Observability + recovery

Agent vs workflow vs automation

Type	Decision control	Typical use
Deterministic workflow	Steps mostly fixed	ETL, payments, scheduled reports
Tool-calling agent	Model chooses among tools	Support assistant, research helper
Graph agent	State + controlled branching	Long-running case handling
Multi-agent system	Specialist coordination	Complex research/support
Autonomous loop	Repeated observe-edit-test	Coding optimization, AVO-style exploration

ReAct style loop

```
GOAL
 -> reason about next action
 -> choose tool
 -> execute
 -> observe result
 -> update state
 -> verify
 -> continue / stop
```

Termination rules

- Max steps / max tool calls
- Time budget
- Cost budget
- Repeated-state detector
- No-progress detector
- Confidence/risk threshold
- Human approval checkpoint
- Hard stop on policy violation

Memory design

- Working memory: current task state
- Conversation memory: relevant thread history
- Semantic memory: durable facts/preferences
- Episodic memory: past task outcomes
- Procedural memory: reusable instructions/skills
- Memory write policy: kya store karna hai, kyun, retention kitna, delete kaise

Tool design - golden rules

72. Tool ka naam specific rakho.
73. Description me side effects clearly likho.
74. Strict schema use karo.
75. Read-only vs write tool separate rakho.
76. Authorization user/tenant context se derive karo.
77. Sensitive action ke liye preview + approval.
78. Timeout + retry + idempotency.
79. Tool output validate aur size-limit karo.
80. Audit record maintain karo.

Reflection / verifier pattern

```
DRAFT -> VERIFY ->
 PASS -> RETURN
 FAIL -> REPAIR -> VERIFY
 REPEAT until max_attempts
```

Agentic RAG

- Agent decide karta hai query rewrite chahiye ya nahi.
- Multiple retrievers invoke ho sakte hain.
- Evidence compare/verify kiya ja sakta hai.
- Low-confidence result par human escalation ho sakta hai.
- Every tool call trace hona chahiye.

Multi-agent patterns

- Supervisor -> specialists -> verifier
- Router -> domain expert
- Parallel experts -> aggregator
- Planner -> workers -> judge
- Researcher -> fact-checker -> writer
- Support router -> order/refund/knowledge specialists

Deep agent concepts - 2026

- Tool use + virtual filesystem
- Context compression/offloading
- Subagent delegation
- Long-running task support
- Human approval
- Sandboxed execution
- Skills and memory
- Streaming task events
- MCP connectivity

Current Deep Agents docs planning, subagents, filesystem tools, context management, memory, human approval aur MCP integration ko built-in agent-harness capabilities ke roop me describe karti hain. LangChain ka current `create_agent` configurable harness hai; LangGraph long-running stateful orchestration ke liye low-level runtime deta hai.

Autonomous coding / AVO lesson

Uploaded AVO paper me agent previous solutions, domain knowledge, tools aur evaluation feedback ke saath iterative planning, implementation, testing, debugging aur adaptation karta hai. Lesson: autonomy tab useful hai jab evaluation objective, rollback

path aur execution environment controlled ho. Source paper ne multi-day autonomous kernel optimization ko example ke roop me demonstrate kiya.

Agentic AI interview Q&A

Sawal - Agent aur chatbot me main difference?

Jawaab - Chatbot mostly conversation handle karta hai. Agent goal pursue karte hue tools, state aur multi-step actions use kar sakta hai.

Level: Basic

Sawal - Agent ko autonomous kab banana chahiye?

Jawaab - Jab task multi-step ho, tool decisions dynamic hon aur safe termination/evaluation possible ho. Simple deterministic work me unnecessary autonomy avoid karo.

Level: Normal

Sawal - Planner aur executor separate kyun?

Jawaab - Planning ko execution se isolate karke traceability, retries aur specialized policies better manage ki ja sakti hain.

Level: Advanced

Sawal - Verifier agent ka role?

Jawaab - Output/tool result ko independent criteria se check karna aur fail hone par repair trigger karna.

Level: Advanced

Sawal - Agent loop infinite kyun hota hai?

Jawaab - No-progress detection, max steps, ambiguous goals, tool failures ya repeated state ki wajah se. Hard budgets lagao.

Level: Normal

Sawal - Memory blindly store karna sahi hai?

Jawaab - Nahi. Sensitive data, stale facts aur wrong preferences future behavior distort kar sakte hain. Retention/write policy chahiye.

Level: Advanced

Sawal - Multi-agent kab avoid karoge?

Jawaab - Jab single agent/workflow same outcome simpler, cheaper aur more testable way se de raha ho.

Level: Advanced

Sawal - Deep agent aur custom graph me trade-off?

Jawaab - Deep agent prebuilt capabilities fast deta hai; custom graph me exact state transitions aur control zyada milta hai.

Level: Advanced

Sawal - Agentic RAG vs fixed RAG?

Jawaab - Fixed RAG predetermined retrieval path use karta hai. Agentic RAG query strategy dynamically adapt kar sakta hai.

Level: Advanced

Sawal - AVO se AI engineer kya seekh sakta hai?

Jawaab - Agent ko candidate generator se iterative variation/evaluation loop tak le ja sakte ho, but objective function, correctness gates and rollback critical hain.

Level: Research

25 ROLE-WISE MEGA INTERVIEW Q&A

Ab answers ko role ke hisaab se bolo: AI Engineer, Data Scientist, Data Analyst aur Data Engineer.

AI Engineer - Interview Set

Sawal - Production RAG me ACL kaise enforce karoge?

Jawaab - Retrieval metadata me tenant/user permissions propagate karunga aur authorization ko retrieval se pehle/baad ke layers me enforce karunga; filtered evidence ke bina answer nahi dena.

Level: Advanced

Sawal - Tool gateway kyun?

Jawaab - Central layer me auth, schema validation, rate limit, logging aur routing ko standardize kar sakte hain.

Level: Advanced

Sawal - Fallback model ka risk?

Jawaab - Smaller model quality lower ho sakti hai; fallback ko same eval suite aur policy checks se validate karna chahiye.

Level: Advanced

Sawal - Agent latency ka bottleneck kaise find?

Jawaab - Trace spans se retrieval, model, tool aur verifier time compare karunga; parallelization/cache/routing se optimize karunga.

Level: Advanced

Sawal - Structured output fail ho to?

Jawaab - Schema validation -> constrained retry/repair -> safe fallback; invalid payload ko side-effecting tool tak nahi bhejna.

Level: Normal

Sawal - LLM provider switch ke liye kya design?

Jawaab - Model adapter/standard interface, capability matrix, prompt compatibility tests aur provider-specific fallbacks.

Level: Advanced

Sawal - Human approval kis action par?

Jawaab - Irreversible, high-impact, high-cost ya permission-sensitive actions par risk-based approval.

Level: Normal

Sawal - Agent observability me kya log?

Jawaab - Request ID, model/version, prompt version, retrieval IDs, tool names/latency, state transitions, policy decisions, outcome and cost-safe metadata.

Level: Advanced

Data Scientist - Interview Set

Sawal - Offline metric improve par business metric same kyun?

Jawaab - Metric misaligned, threshold wrong, selection bias, latency, intervention effect ya user behavior change ho sakta hai.

Level: Advanced

Sawal - Model calibration kab important?

Jawaab - Risk scores, ranking by probability, triage aur decision thresholds jahan probability ka meaning operational action se linked ho.

Level: Advanced

Sawal - Experiment me novelty effect kya?

Jawaab - Users nayi feature se temporarily different behave kar sakte hain; experiment duration aur cohort stabilization consider karo.

Level: Normal

Sawal - Causal claim ke liye best evidence?

Jawaab - Randomized experiment ideal case ho sakta hai; observational settings me identification assumptions explicit honi chahiye.

Level: Advanced

Sawal - Imbalanced classification me threshold kaise choose?

Jawaab - Business cost, precision-recall tradeoff, calibration aur review capacity ko together consider karo.

Level: Advanced

Sawal - Time-series cross validation?

Jawaab - Rolling/expanding windows ya forward-chaining use karo so future information leak na ho.

Level: Advanced

Sawal - Feature store kyu?

Jawaab - Training aur online serving ke beech consistent, point-in-time-correct feature access manage karne me help karta hai.

Level: Advanced

Data Analyst - Interview Set

Sawal - Metric definition me grain kya?

Jawaab - Ek row kis business entity/time unit ko represent karti hai; grain wrong hua to aggregates double-count ho sakte hain.

Level: Advanced

Sawal - Cohort retention ka denominator?

Jawaab - Har cohort ke baseline population ko clearly define karke later periods me active users divided by that cohort baseline.

Level: Normal

Sawal - Dashboard alert kab create karoge?

Jawaab - Jab deviation meaningful ho, action clear ho aur noise manageable ho.

Level: Normal

Sawal - SQL dedup ka common pattern?

Jawaab - ROW_NUMBER over business key ordered by event/update timestamp, then rn = 1.

Level: Normal

Sawal - Analyst AI copilot ko reliable kaise?

Jawaab - Semantic schema, read-only access, SQL validation, query limits, result checks and cited metric definitions.

Level: Advanced

Sawal - Correlation high hai to action loge?

Jawaab - Correlation hypothesis generate karta hai; business action se pehle confounders aur causal evidence check karunga.

Level: Normal

Data Engineer - Interview Set

Sawal - Backfill aur replay me difference?

Jawaab - Backfill historical time range ko recompute karta hai; replay raw/event log se processing dobara chalata hai. Dono ke liye idempotency important hai.

Level: Advanced

Sawal - At-least-once consumer me duplicates kaise?

Jawaab - Message retry se duplicate delivery ho sakti hai; idempotent sink/dedup key use karo.
Level: Normal

Sawal - Watermark too aggressive ho to?

Jawaab - Late valid events drop ho sakte hain. Delay threshold ko observed lateness aur business tolerance ke basis par tune karo.
Level: Advanced

Sawal - Partition key kaise choose?

Jawaab - Even distribution + query locality + ordering requirement + hot-key risk ko together dekho.
Level: Advanced

Sawal - Schema breaking change ka safe plan?

Jawaab - Versioned contract, compatibility window, dual-read/write where needed, validation and rollback.
Level: Advanced

Sawal - Data quality gate ka blast radius kaise control?

Jawaab - Critical vs advisory checks separate, quarantine bad partitions, notify owners, and prevent only unsafe downstream publishes.
Level: Advanced

Sawal - Lakehouse maintenance kyun?

Jawaab - Snapshots/files/metadata accumulate ho sakte hain; retention and compaction health ko maintain rakhte hain.
Level: Normal

26 30-DAY REVISION + ROLE CHECKLISTS

Har din concept + code + interview + project ka combination rakho. Goal sirf padhna nahi, explain aur build karna hai.

Days	Focus	Output
Day 1-3	**LLM + Transformer**	Tokenization, embeddings, attention, decoding; 15 Q aloud
Day 4-6	**Prompt + Context**	Structured output, injection, context budgeting; 2 labs
Day 7-9	**RAG**	Chunking, hybrid retrieval, reranking, ACL; mini RAG design
Day 10-12	**LangChain + LangGraph**	Agent harness, tools, middleware, state, persistence; build one agent
Day 13-15	**Agentic AI + MCP**	Planning, memory, HITL, multi-agent, MCP; draw architecture
Day 16-18	**Data Science**	Stats, experiments, ML metrics, leakage, time series; solve 20 Q
Day 19-21	**Data Analytics**	Advanced SQL, KPI, cohort, funnel, dashboard storytelling; 15 SQL problems
Day 22-24	**Data Engineering**	Kafka, Spark, CDC, lakehouse, orchestration, quality; design a pipeline
Day 25-27	**MLOps + Security**	Eval, tracing, drift, canary, rate limits, prompt injection, audit
Day 28	**System Design**	One end-to-end AI + data architecture on paper
Day 29	**Project Defense**	90-sec intro + 10 deep follow-ups + failure modes
Day 30	**Mock Interview**	40 mixed questions + coding + system design + post-mortem

AI Engineer must-know checklist

[] LLM internals
[] Transformers + attention
[] Prompt/context engineering
[] Structured outputs
[] RAG
[] Hybrid retrieval
[] Reranking
[] ACL-aware retrieval
[] LangChain
[] LangGraph
[] Deep Agents
[] Tool calling
[] MCP
[] Memory
[] Human-in-the-loop
[] Multi-agent patterns
[] Evaluation
[] Observability
[] Security
[] System design
[] Cost/latency
[] Deployment
[] Incident recovery

Data Scientist must-know checklist

[] EDA
[] Probability
[] Confidence intervals
[] Hypothesis tests
[] Experiment design
[] Leakage
[] Cross-validation
[] Metrics
[] Calibration
[] Time series
[] Recommendations
[] NLP
[] Business impact

Data Analyst must-know checklist

[] Advanced SQL
[] Joins/grain
[] CTE
[] Window functions
[] Cohorts
[] Funnels
[] Retention
[] KPI contracts
[] Dashboard design
[] Storytelling
[] Experiment readout

Data Engineer must-know checklist

[] ETL/ELT
[] Warehouse/lakehouse
[] CDC
[] Kafka
[] Spark
[] Event time
[] Watermarks
[] Backpressure
[] Airflow
[] dbt
[] Data contracts
[] Data quality
[] Lineage
[] Feature store
[] Backfill
[] Idempotency
[] Schema evolution

27 END-TO-END IMPLEMENTATION PLAYBOOK

Ab theory ko project me kaise lagana hai: architecture, repo structure, milestones, tests, evaluation aur production checklist ek hi jagah.

Project 01 - Enterprise RAG + Support Agent

```
USER -> API -> AUTH
    -> QUERY ROUTER
    -> RETRIEVAL
     |-- keyword search
     |-- vector search
     |-- metadata / ACL filters
     -> RERANK
    -> AGENT
     |-- knowledge tool
     |-- ticket tool
     |-- order tool
    -> VERIFIER
    -> HUMAN APPROVAL (risk based)
    -> RESPONSE + CITATIONS
    -> TRACE / EVAL
```

Build order

81. 20-50 representative documents collect karo.
82. Metadata contract define karo: source, tenant, page, timestamp, ACL.
83. Chunk + embed + index pipeline banao.
84. Hybrid retrieval + reranking add karo.
85. Read-only tools first add karo.
86. Agent + verifier add karo.
87. Write actions ko approval gate ke peeche rakho.
88. Golden eval set aur adversarial cases banao.
89. Tracing + cost + latency dashboards add karo.
90. Load/failure tests ke baad deployment karo.

Definition of Done

- Correct answer + source citation
- Unauthorized document never retrieved
- Tool schema validation
- Timeout + retry + idempotency
- 20+ golden eval cases
- Prompt injection cases tested
- Trace IDs available
- Cost per task tracked
- Rollback path documented

Project 02 - AI Analytics Copilot

```
USER QUESTION
 -> INTENT CLASSIFIER
 -> SEMANTIC METRIC LAYER
 -> READ-ONLY SQL GENERATOR
 -> SQL VALIDATOR
 -> DRY RUN / EXPLAIN
 -> EXECUTE WITH LIMITS
 -> RESULT VALIDATOR
 -> CHART SPEC
 -> NARRATIVE
 -> AUDIT LOG
```

Safe SQL policy

- Only SELECT/approved read operations

- Allowlisted schemas/tables
- Blocked dangerous keywords
- Maximum scan/row/time budget
- PII columns masked or denied
- Result-size limit
- Query and metric definition logged
- Human review for high-impact exports

Project 03 - Agentic DataOps Assistant

```
EVENT: pipeline_failed
   |
   v
TRIAGE AGENT
 |--> log tool
 |--> lineage tool
 |--> schema tool
 |--> recent-change tool
   |
DIAGNOSIS
 |--> deterministic checks
 |--> evidence summary
   |
REPAIR PLAN
 |--> safe auto-fix OR human approval
   |
VERIFY
 |--> data-quality checks
 |--> smoke query
   |
RESOLVE / ESCALATE
```

Agent ko kya allow karna hai?

- Read logs, metadata, lineage, test results
- Propose remediation
- Create patch/change request
- Run safe validation in sandbox
- Never directly delete production data
- High-risk actions require explicit approval

Project 04 - Streaming Anomaly + Incident Agent

```
EVENTS -> KAFKA -> STREAM PROCESSOR
    -> VALIDATE / DEDUP / WATERMARK
    -> FEATURE AGGREGATES
    -> ANOMALY DETECTOR
    -> ALERT ROUTER
    -> INCIDENT AGENT
    -> RUNBOOK / HUMAN
    -> RESOLUTION + POSTMORTEM
```

Streaming checks

- Late-event rate
- Duplicate rate
- Partition lag
- Throughput
- Window completeness
- State growth
- Dead-letter volume
- Alert precision

API + Integration Engineering - AI Engineer ke liye

- REST: resource-oriented endpoints, status codes, pagination, filtering

- OAuth/OIDC basics: identity, access token, scopes, expiry
- Webhook: event-driven callback; signature verify karo
- Idempotency key: duplicate request se duplicate business effect avoid karo
- Retries: sirf transient failures par; exponential backoff + jitter
- Timeouts: every upstream call ka bounded timeout
- Rate limits: per-user/tenant/provider budget
- Schema versioning: backward-compatible contracts
- Queues: async work, backpressure, dead-letter handling

Webhook verification example

```
POST /webhook
X-Signature: hmac-sha256=...

BODY -> compute HMAC -> constant-time compare
   -> verify timestamp / replay window
   -> parse schema
   -> dedupe event_id
   -> enqueue work
```

Cloud + deployment checklist

- Container image immutable tag ke saath build karo.
- Secrets source code me nahi; secret manager/env injection use karo.
- Health + readiness endpoints rakho.
- Horizontal scaling aur queue-based workers plan karo.
- Object storage ko durable artifacts/corpus ke liye use karo.
- Database backups aur restore drill define karo.
- Infrastructure changes version-controlled rakho.
- CI/CD me tests, security scan, deployment gate aur rollback rakho.

Production readiness matrix

Dimension	Must-have
Correctness	Golden eval + regression set + domain review
Safety	Authz + policy + prompt injection tests + approval
Reliability	Timeout + retry + circuit breaker + idempotency
Scale	Queue + autoscaling + load test + backpressure
Data	Quality checks + lineage + freshness + contracts
Observability	Trace + metrics + logs + alerts
Cost	Token/tool budget + caching + model routing
Recovery	Backup + rollback + RPO/RTO + incident runbook

Repo structure idea

```
apps/
 api/
 worker/
 agent/
services/
 retrieval/
 tools/
 evaluation/
 policy/
data/
 contracts/
 schemas/
infra/
 docker/
 k8s/
 terraform/
evals/
 golden/
 adversarial/
 regression/
observability/
 traces/
 dashboards/
```

Project documentation me kya likhna hai?

- Problem statement
- Architecture diagram
- Data flow
- Tool catalog
- Permission matrix
- Evaluation methodology
- Failure modes
- Latency/cost budget
- Security controls
- Deployment + rollback plan
- Known limitations
- Future improvements

Demo sequence - interview ke liye

91. 10 sec: problem + user.
92. 20 sec: architecture.
93. 20 sec: live happy path.
94. 15 sec: ek failure/guardrail case.
95. 15 sec: evaluation metrics.
96. 15 sec: architecture trade-off.
97. 15 sec: tumne personally kya build kiya.

Golden rule

Interview me sirf "maine agent bana diya" mat bolo. Explain karo ki agent ko kya decision lena diya, kya deterministic rakha, tool permissions kya hain, quality kaise measure ki aur failure par system kya karta hai.

29 ADVANCED SQL + DATA ANALYSIS MASTERCLASS

Data Analyst, Data Scientist aur AI Engineer ke liye SQL ko sirf query language nahi, business reasoning tool ki tarah use karo.

Data analysis ka real workflow

98. Business question ko measurable metric me convert karo.
99. Grain define karo: ek row exactly kya represent karti hai?
100. Source tables aur ownership identify karo.
101. SQL se raw data ko validate karo.
102. EDA ke liye distribution, nulls, duplicates, outliers aur segments check karo.
103. Hypothesis banao aur experiment/analysis design karo.
104. Result ko business context ke saath interpret karo.
105. Dashboard/brief me action, limitation aur next step clearly bolo.

Advanced SQL patterns

Dedup + rolling metric [SQL]

```
WITH cleaned AS (
 SELECT
  customer_id,
  DATE(order_ts) AS order_date,
  revenue,
  ROW_NUMBER() OVER (
   PARTITION BY order_id
   ORDER BY updated_at DESC
  ) AS rn
 FROM orders
), base AS (
 SELECT * FROM cleaned WHERE rn = 1
)
SELECT
 order_date,
 SUM(revenue) AS daily_revenue,
 SUM(SUM(revenue)) OVER (
  ORDER BY order_date
  ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
 ) AS rolling_7d_revenue
FROM base
GROUP BY order_date
ORDER BY order_date;
```

Cohort retention SQL

Cohort base table [SQL]

```
WITH first_seen AS (
 SELECT customer_id, MIN(DATE(event_ts)) AS cohort_date
 FROM events
 GROUP BY customer_id
), activity AS (
 SELECT DISTINCT
  e.customer_id,
  f.cohort_date,
  DATE_TRUNC('month', e.event_ts) AS activity_month
 FROM events e
 JOIN first_seen f USING (customer_id)
), cohort AS (
 SELECT
  cohort_date,
  activity_month,
  COUNT(DISTINCT customer_id) AS active_users
 FROM activity
 GROUP BY 1,2
)
SELECT * FROM cohort ORDER BY cohort_date, activity_month;
```

Funnel analysis

Funnel me har step ka denominator clear rakho. Example: Visit -> Signup -> Activation -> Paid. Har transition par conversion aur drop-off dono report karo. Sirf final conversion dekhne se bottleneck hide ho sakta hai.

Funnel snapshot [SQL]

```
SELECT
 COUNT(DISTINCT CASE WHEN event = 'visit' THEN user_id END) AS visits,
 COUNT(DISTINCT CASE WHEN event = 'signup' THEN user_id END) AS signups,
 COUNT(DISTINCT CASE WHEN event = 'activation' THEN user_id END) AS activations,
 COUNT(DISTINCT CASE WHEN event = 'paid' THEN user_id END) AS paid
FROM product_events
WHERE event_date >= CURRENT_DATE - INTERVAL '30' DAY;
```

Data quality checks - analysis se pehle

- Uniqueness: business key duplicate to nahi?
- Completeness: critical fields me null rate kitna?
- Validity: values allowed domain ke andar hain?
- Consistency: same metric alag sources me same meaning rakhta hai?
- Freshness: latest partition expected time par available hai?
- Distribution: sudden shift ya impossible spike to nahi?
- Referential integrity: foreign keys / entity IDs resolve ho rahe hain?

Statistics jo practical interview me explain karni hain

- Mean, median, mode, variance, standard deviation
- Percentile, quartile, IQR, outlier reasoning
- Probability aur conditional probability
- Confidence interval aur margin of error
- Null/alternative hypothesis
- p-value aur significance level
- Type I / Type II error
- Two-sample aur paired comparison
- Chi-square for categorical relationships
- Correlation vs causation
- Regression ka interpretation
- Sampling bias aur confounding

Data Analyst interview Q&A

Sawal - Grain kya hota hai?

Jawaab - Ek row kis business event/entity/time level ko represent karti hai. Grain galat hua to joins ke baad double counting ho sakti hai.

Level: Basic

Sawal - LEFT JOIN se rows double kyun hoti hain?

Jawaab - Agar right table me join key par multiple matches hain to one-to-many multiplication hoti hai. Pehle grain aur key uniqueness verify karo.

Level: Normal

Sawal - Window function kab use karoge?

Jawaab - Row context preserve rakhte hue ranking, running totals, lag/lead aur rolling metrics ke liye.

Level: Normal

Sawal - CTE kyun useful hai?

Jawaab - Complex query ko logical stages me todne, readability badhane aur intermediate logic validate karne ke liye.

Level: Basic

Sawal - Cohort analysis ka purpose?

Jawaab - Same starting event/date wale users ko group karke time ke saath retention/behavior compare karna.
Level: Normal

Sawal - Retention numerator/denominator kaise define?

Jawaab - Numerator active users in a later period; denominator original cohort population. Dono definitions explicit honi chahiye.
Level: Normal

Sawal - Dashboard me 100 metrics rakhna sahi hai?

Jawaab - Usually nahi. Role-specific decision metrics, supporting diagnostics aur clear ownership better hote hain.
Level: Normal

Sawal - A/B test me primary metric kyun define karte?

Jawaab - Taaki success criteria experiment ke baad opportunistically change na ho. Guardrail metrics bhi predefine karo.
Level: Advanced

Sawal - Correlation high hai to causation?

Jawaab - Nahi. Confounders ya reverse causality possible hai. Causal claim ke liye stronger design/evidence chahiye.
Level: Basic

Sawal - Data storytelling ka structure?

Jawaab - Context -> key finding -> evidence -> business impact -> action -> limitation/next step.
Level: Normal

30 DATA SCIENTIST KA DEEP MASTERCLASS

Statistics + ML + experimentation + production thinking ko combine karo; sirf model train karna Data Science nahi hai.

DS lifecycle

106. Problem framing
107. Data acquisition + validation
108. EDA + hypothesis formation
109. Feature engineering
110. Baseline model
111. Cross-validation / temporal validation
112. Metric selection
113. Error analysis
114. Business simulation
115. Deployment plan
116. Monitoring + retraining policy

Modeling decisions

- Classification: probability + threshold + cost matrix
- Regression: continuous outcome; residual analysis important
- Clustering: segmentation, not ground truth
- Ranking/recommendation: ordering quality + business constraints
- Time series: temporal splits; leakage risk high
- NLP: task-specific representation, retrieval, fine-tuning or LLM depending on requirement
- Anomaly detection: define what counts as a useful alert before selecting model

Feature engineering examples

Feature pipeline idea [FEATURE]

```
raw_event_ts
 -> hour_of_day
 -> day_of_week
 -> recency_days
 -> frequency_30d
 -> monetary_90d
 -> rolling_mean_7d
 -> trend_slope
 -> missingness_flag
```

Leakage examples

- Future outcome used as feature
- Post-event status accidentally included
- Target-derived aggregate computed using full dataset
- Random split on time-dependent data
- Normalization/scaling fit on full dataset before split
- Duplicate users/entities across train and test

Imbalanced classification

- Accuracy misleading ho sakti hai.
- Precision-recall curve inspect karo.
- Threshold business cost ke according tune karo.
- Class weights / resampling sirf validation design ke saath use karo.
- Calibration check karo agar probability operational decision me use hogi.

Experiment design master checklist

- Randomization unit
- Primary metric
- Guardrail metrics
- Power / sample size
- Duration
- Exposure eligibility
- Interference / spillover
- Novelty effects
- Missing data
- Stopping rule
- Segment analysis
- Post-experiment interpretation

Time-series checklist

- Train on past, validate on future
- Seasonality and holidays model karo
- Rolling features past-only rakho
- Forecast horizon define karo
- Backtesting slices use karo
- Prediction intervals/report uncertainty where relevant

Data Scientist interview Q&A - advanced

Sawal - Baseline model kyun?

Jawaab - Baseline tumhe reference deta hai. Complex model tabhi useful hai jab measurable improvement aaye aur cost/latency acceptable rahe.

Level: Basic

Sawal - Cross-validation aur time-series split me difference?

Jawaab - Random CV iid-like data me useful ho sakta hai; temporal data me future leakage avoid karne ke liye forward/rolling split better hai.

Level: Advanced

Sawal - Calibration kya hai?

Jawaab - Predicted probability aur observed frequency ke alignment ko calibration kehte hain. Risk decisions me important ho sakta hai.

Level: Advanced

Sawal - Feature importance ko causal effect samajhna sahi hai?

Jawaab - Nahi. Feature importance predictive contribution dikha sakta hai, causal effect nahi.

Level: Advanced

Sawal - PR-AUC kab useful?

Jawaab - Rare-positive classification me precision-recall tradeoff ko summarize karne ke liye useful ho sakta hai.

Level: Normal

Sawal - Model score improve but production worsen kyun?

Jawaab - Distribution shift, latency, missing features, threshold mismatch, feedback loop ya metric mismatch ho sakta hai.

Level: Advanced

Sawal - Drift ke types?

Jawaab - Input/data distribution shift, target/relationship shift aur model performance degradation ko alag monitor karo.
Level: Advanced

Sawal - Retraining kab?

Jawaab - Fixed schedule ke saath blindly nahi. Drift, performance degradation, data freshness aur business change ke signals ke basis par policy define karo.
Level: Advanced

Sawal - Causal inference ka basic mindset?

Jawaab - Counterfactual socho: intervention na hota to outcome kya hota? Design aur assumptions explicit rakho.
Level: Advanced

Sawal - Error analysis ka best output?

Jawaab - Errors ko meaningful slices me group karke top failure modes, examples aur next engineering/model changes identify karo.
Level: Normal

31 DATA ENGINEER KA DEEP MASTERCLASS

Reliable data supply chain banao: ingest -> validate -> transform -> serve -> observe -> recover.

Architecture layers

```
SOURCES
 | DB / SaaS / APIs / files / events
 v
INGESTION
 | batch / CDC / streaming
 v
RAW / BRONZE
 | validation + schema
 v
CURATED / SILVER
 | joins + business rules
 v
SERVING / GOLD
 | BI / ML / AI / APIs
 v
OBSERVABILITY
 | freshness / volume / quality / lineage
```

Warehouse vs Lake vs Lakehouse

Architecture	Kaam	Typical use
Warehouse	Curated analytics + SQL serving	BI, reporting, semantic models
Data Lake	Flexible raw/large-scale storage	Raw logs, files, multimodal corpus
Lakehouse	Lake storage + table transactions/governance	Unified analytics, ML, batch + streaming

Data modeling

- Star schema: fact + dimensions
- Snowflake schema: normalized dimensions
- Slowly changing dimensions: current vs historical state
- Surrogate key vs natural key
- Business grain + uniqueness constraints
- Partitioning/clustering strategy
- Semantic layer: metric definitions + ownership

CDC + idempotency

```
source_change
 -> event_id
 -> op = INSERT | UPDATE | DELETE
 -> event_ts
 -> source_version
 -> downstream MERGE/UPSERT
 -> dedupe on event_id
 -> checkpoint / offset commit
```

Airflow thinking

Airflow ke current docs ke mutabik DAG workflow ko model karta hai; tasks discrete units hain aur dependencies order define karti hain. Retry, timeout, backfill aur re-run semantics ko production design ka hissa samjho.

Data Engineer interview Q&A - advanced

Sawal - ETL vs ELT?

Jawaab - ETL me transform load se pehle; ELT me raw data ko platform par load karke transformation downstream hota hai.

Level: Basic

Sawal - Idempotent pipeline?

Jawaab - Same input ko retry karne par same intended final state mile aur duplicate side effects na bane.
Level: Normal

Sawal - CDC me delete kaise?

Jawaab - Delete event/tombstone capture karke downstream policy ke hisaab se soft ya hard delete apply karo.
Level: Normal

Sawal - Partition key kaise choose?

Jawaab - Even distribution, query locality, ordering need aur hot-key risk together evaluate karo.
Level: Advanced

Sawal - Small files problem?

Jawaab - Too many tiny files metadata/scan overhead badhate hain; compaction aur write sizing strategy help karti hai.
Level: Normal

Sawal - Backfill safe kaise?

Jawaab - Versioned input, partition-scoped rerun, idempotent transform, isolated output, validation aur downstream impact review.
Level: Advanced

Sawal - Data contract kya?

Jawaab - Schema + field meaning + owner + freshness SLA + compatibility/change policy ka explicit agreement.
Level: Advanced

Sawal - Data lineage kyun?

Jawaab - Upstream source se downstream dashboard/model tak impact analysis, debugging aur governance ke liye.
Level: Normal

Sawal - Schema evolution ka safe plan?

Jawaab - Backward-compatible change, versioned contract, migration window, validation aur rollback.
Level: Advanced

Sawal - Exactly-once ko kaise discuss?

Jawaab - Transport-level claim se zyada end-to-end semantics dekho: replay-safe processing, transactional sink ya idempotent writes aur checkpointing.
Level: Advanced

32 STREAMING + BIG DATA ENGINEERING

Kafka + stream processing + event-time reasoning AI systems ke live data backbone ko samajhne ke liye important hai.

Streaming mental model

```
PRODUCER -> TOPIC -> PARTITIONS -> CONSUMER GROUP
             |
             +-> STREAM PROCESSOR
               |-> WINDOW AGGREGATE
               |-> STATE STORE
               |-> SINK / ALERT / FEATURE
```

Event time vs processing time

- Event time = event actually kab hua.
- Processing time = system ne kab process kiya.
- Late data ko handle karne ke liye watermarking aur allowed lateness strategy important hai.
- Wrong watermark se late valid events drop ya state cost high ho sakti hai.

Kafka concepts

- Topic, partition, offset
- Producer acknowledgements
- Consumer group
- Partition assignment
- Ordering guarantee within partition
- Retention
- Replay
- Dead-letter topic
- Lag monitoring

Spark / stream design

- Micro-batch ya continuous-style processing capability workload ke according choose karo.
- Stateful windows ka memory impact samjho.
- Watermark + dedup combine karo.
- Checkpoint/restart behavior test karo.
- Late-event metrics monitor karo.

Lakehouse practicals

- ACID transactions
- Schema enforcement
- Schema evolution
- Time travel
- Batch + streaming same table pattern
- MERGE/upsert for CDC
- Compaction / file layout
- Partition pruning

Streaming anomaly example

Streaming rule idea [SQL]

```
SELECT
 customer_id,
 window_start,
```

```
 SUM(amount) AS spend_10m,
 COUNT(*) AS txns_10m
FROM transactions
GROUP BY customer_id, window_start
HAVING SUM(amount) > 100000
  OR COUNT(*) > 1000;
```

Interview line

Streaming me main sirf throughput nahi dekhta; event-time correctness, duplicate handling, state growth, lag, backpressure aur recovery semantics bhi define karta hoon.

Big Data interview Q&A

Sawal - Consumer lag kya?

Jawaab - Latest available offset aur consumer progress ke beech ka gap; sustained lag capacity/processing issue signal ho sakta hai.

Level: Basic

Sawal - Hot partition kya?

Jawaab - Uneven key distribution se ek partition disproportionate load le leta hai.

Level: Advanced

Sawal - Watermark ka trade-off?

Jawaab - Late data tolerance badhao to state zyada retain ho sakta hai; aggressive watermark memory bachata hai but valid late events drop kar sakta hai.

Level: Advanced

Sawal - Backpressure kya?

Jawaab - Downstream slow hone par upstream processing ko control karna taaki unbounded queue/memory growth na ho.

Level: Normal

Sawal - Replayable pipeline kyun?

Jawaab - Historical bug fix, recovery aur backfill ke liye raw event log se deterministic recomputation useful hoti hai.

Level: Normal

Sawal - Dead-letter queue/topic ka use?

Jawaab - Poison messages ko isolate karna, main stream ko block na hone dena aur later repair/replay allow karna.

Level: Normal

33 AGENTIC AI + LANGCHAIN IMPLEMENTATION LAB

Current LangChain docs ke `create_agent` harness ko base bana kar tool-calling agent ka practical flow samjho. Framework version project setup ke time re-check karo.

Current LangChain agent pattern

LangChain agent skeleton [TS]

```ts
import { createAgent, tool } from "langchain";
import * as z from "zod";

const lookupOrder = tool(
 async ({ orderId }) => ({ orderId, status: "shipped" }),
 {
  name: "lookup_order",
  description: "Approved order status lookup",
  schema: z.object({ orderId: z.string() })
 }
);

const agent = createAgent({
 model: process.env.MODEL_ID,
 tools: [lookupOrder],
 systemPrompt: "You are a support agent. Never invent order data. Use the tool for live status."
});

const result = await agent.invoke({
 messages: [{ role: "user", content: "Mera order ORD-101 kaha hai?" }]
});
```

Is code me kya ho raha hai?

117. Tool ka contract schema se define hua.
118. Agent model ko tool choose karne deta hai.
119. Live state ke liye tool use hota hai; model memory se guess nahi karta.
120. System instruction tool usage policy set karti hai.
121. Production me tool authorization, timeout, audit aur output validation add karni hai.

Tool safety wrapper

```ts
async function safeToolCall(input, ctx) {
 validateTenant(ctx.tenantId);
 validateSchema(input);
 const result = await withTimeout(runTool(input), 3000);
 return sanitizeToolResult(result, { maxChars: 6000 });
}
```

Agent budget

```
MAX_STEPS=8
MAX_TOOL_CALLS=6
MAX_WALL_TIME_MS=30000
MAX_SPEND_USD=0.05

stop when:
- goal verified
- policy denied
- budget exceeded
- repeated state detected
- human approval required
```

Agent evaluation dataset

- Happy-path tasks
- Ambiguous requests
- Tool failure cases
- Wrong/empty tool results
- Prompt injection attempts
- Permission boundary cases

- High-risk write requests
- Long-horizon tasks
- No-progress loops
- Multi-agent conflicts

Agentic AI practice Q&A

Sawal - Agent ko workflow se kab replace karoge?

Jawaab - Jab steps deterministic hon aur decision branching fixed ho. Har problem me agent add karna complexity badha sakta hai.

Level: Basic

Sawal - Agentic system me deterministic layer kyun?

Jawaab - Auth, billing rules, schema validation, risk checks aur irreversible actions ko deterministic boundary dena reliability improve karta hai.

Level: Advanced

Sawal - Tool description kitni important?

Jawaab - Bahut important. Agent tool selection ke liye description/schema par depend karta hai; ambiguous descriptions wrong tool calls kara sakti hain.

Level: Normal

Sawal - Agent ko production me unrestricted access dena?

Jawaab - Avoid. Least privilege, scoped credentials, per-tool permissions aur approval gates use karo.

Level: Security

Sawal - Agent eval me sirf final answer enough?

Jawaab - Nahi. Tool selection, arguments, retrieval, policy compliance, latency, cost aur final outcome bhi evaluate karo.

Level: Advanced

34 LANGGRAPH + MCP + MULTI-AGENT LAB

Long-running stateful workflows, human approval aur external tool connectivity ko controlled graph me combine karo.

LangGraph design pattern

```
START
 |
 v
CLASSIFY -> RETRIEVE -> DRAFT
   |     |     |
   |     v     v
   |    EVIDENCE -> VERIFY
   |           |
   +-------------------+
         |
    PASS -> RESPOND
    FAIL -> REPAIR
    HIGH-RISK -> HUMAN
```

State contract example

```
type State = {
 userId: string;
 task: string;
 evidence: string[];
 draft: string;
 verification: "pending" | "pass" | "fail";
 approval: "not_needed" | "pending" | "approved" | "rejected";
 traceId: string;
};
```

MCP ka mental model

MCP AI application aur external systems ke beech standard connectivity layer hai. Current TypeScript SDK v2 2026-07-28 specification line ko implement karta hai aur server tools/resources/prompts expose kar sakta hai.

MCP server skeleton [TS]

```
import { McpServer } from "@modelcontextprotocol/server";
import { serveStdio } from "@modelcontextprotocol/server/stdio";
import * as z from "zod/v4";

serveStdio(() => {
 const server = new McpServer({ name: "support-tools", version: "1.0.0" });

 server.registerTool(
  "get_ticket",
  {
   title: "Ticket lookup",
   description: "Read-only ticket lookup",
   inputSchema: { ticketId: z.string() }
  },
  async ({ ticketId }) => ({
   content: [{ type: "text", text: JSON.stringify({ ticketId, status: "open" }) }]
  })
 );

 return server;
});
```

Multi-agent pattern

```
SUPERVISOR
 |--> RESEARCH AGENT ----> EVIDENCE
 |--> ANALYTICS AGENT ----> METRICS
 |--> DOMAIN AGENT ------> DOMAIN ANSWER
 |--> SECURITY AGENT ----> POLICY CHECK
        |
        v
      FINAL VERIFIER
        |
      HUMAN / USER
```

Multi-agent anti-patterns

- Har micro-step ke liye alag agent
- Shared mutable state without ownership
- Agent-to-agent free-form prompts with no schema
- No timeout/deadline
- No final verifier
- No provenance
- Circular delegation
- Duplicate work between specialists

LangGraph/MCP interview Q&A

Sawal - LangGraph kyun?

Jawaab - Jab stateful, branching, retry, persistence, human approval aur long-running workflow ko explicit control chahiye.

Level: Normal

Sawal - Graph node me kya hona chahiye?

Jawaab - Single responsibility step: model call, retrieval, tool execution, validation ya routing. Shared state typed aur minimal rakho.

Level: Advanced

Sawal - MCP aur normal REST API me difference?

Jawaab - REST API custom app-to-app integration hai; MCP AI applications ko tools/resources/prompts discoverable standard ke through expose karne ke liye protocol layer deta hai.

Level: Advanced

Sawal - MCP tool ko trusted maana?

Jawaab - Nahi. Tool metadata aur results ko security boundary ke through validate karo; permissions explicit rakho.

Level: Security

Sawal - Multi-agent se quality automatically improve?

Jawaab - Nahi. More agents cost/latency aur coordination errors bhi badha sakte hain; benchmark ke basis par justify karo.

Level: Advanced

35 PRODUCTION AI SYSTEM DESIGN + SECURITY

System ko happy-path demo se production-grade service tak le jao: reliability, security, observability, scale, cost aur recovery.

Production architecture

```
CLIENT
 |
API GATEWAY -> AUTHN/AUTHZ -> RATE LIMIT
 |
ORCHESTRATOR / AGENT RUNTIME
 |---- MODEL ROUTER
 |---- RETRIEVAL / SEARCH
 |---- TOOL GATEWAY
 |---- STATE / MEMORY
 |---- POLICY / GUARDRAILS
 |---- HUMAN APPROVAL
 |
EVAL + TRACE + METRICS + LOGS
 |
QUEUE / WORKERS / AUTOSCALING
 |
DATA + MODEL + PROMPT VERSION STORE
```

Reliability checklist

- Timeout every upstream call
- Retry only safe transient failures
- Exponential backoff + jitter
- Circuit breaker for repeated upstream failures
- Idempotency for side effects
- Dead-letter for poison jobs
- Graceful degradation
- Health/readiness checks
- Rollback and migration plan
- RTO/RPO defined

AI security - 2025/2026 awareness

- Prompt injection
- Sensitive information disclosure
- Supply-chain risk
- Data/model poisoning
- Improper output handling
- Excessive agency
- System prompt leakage
- Vector/embedding weaknesses
- Misinformation
- Unbounded consumption

OWASP ke 2025 LLM risk list me prompt injection, sensitive information disclosure, supply chain, data/model poisoning, improper output handling, excessive agency, system prompt leakage, vector/embedding weaknesses, misinformation aur unbounded consumption included hain.

Least-privilege agent policy

```
TOOL: refund_order
permission: support.refund.request
conditions:
 - user_authenticated == true
 - order_owner == user_id
 - amount <= policy_limit
```

```
  - no_duplicate_request == true
  - human_approval == true
audit:
  log(request_id, tool, actor, policy_decision, outcome)
```

NIST-style risk management mindset

NIST ka GenAI profile trustworthy/responsible AI risk management ko lifecycle ke across organize karne ka framework deta hai. Practical system me risk identify -> measure -> mitigate -> monitor ka loop rakho.

Cost engineering

- Model routing by task complexity
- Context trimming/compression
- Caching
- Batching where appropriate
- Tool result size limits
- Async processing
- Smaller model for simple tasks
- Per-run budgets
- Per-tenant quotas
- Cost dashboards + anomaly alerts

Incident playbook

122. Detect: alert + trace ID.
123. Contain: disable risky tool or route to read-only mode.
124. Diagnose: inspect model/tool/retrieval/data signals.
125. Recover: rollback prompt/model/index if required.
126. Verify: run smoke + golden eval.
127. Communicate: impact, scope, next update.
128. Learn: root cause + prevention test.

Production system design Q&A

Sawal - P99 latency kya batata hai?

Jawaab - Worst-tail user experience ko represent karne ke liye useful percentile hai; average latency alone spikes hide kar sakti hai.

Level: Normal

Sawal - Retry storm kaise stop?

Jawaab - Central retry budget, jitter, circuit breaker aur upstream-aware retries.

Level: Advanced

Sawal - Circuit breaker kab?

Jawaab - Jab downstream repeated fail ho raha ho aur repeated requests system ko aur pressure de rahe hon.

Level: Normal

Sawal - Graceful degradation example?

Jawaab - Vector DB unavailable ho to approved lexical/cached source se read-only answer ya controlled fallback; unsafe action block.

Level: Advanced

Sawal - RTO vs RPO?

Jawaab - RTO target recovery time; RPO acceptable data-loss window.

Level: Normal

Sawal - AI audit trail me kya?

Jawaab - Request/trace ID, actor, policy decision, retrieval IDs, tool calls, approvals, model/prompt versions aur outcome ka safe metadata.

Level: Advanced

36 MEGA INTERVIEW + 60-DAY MASTER PLAN

Interview ke liye concept, coding, system design aur project defense ko parallel chalao.

40 scenario questions - rapid practice

Sawal - Production RAG me unauthorized document retrieve ho gaya - root cause kaise find karoge?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Agent same tool ko baar-baar call kar raha hai - control kya lagao?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - LLM output valid JSON nahi de raha - recovery path?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - RAG recall high hai lekin answer grounded nahi - next debugging step?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Provider outage me AI assistant ko kaise degrade karoge?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Query latency suddenly 2x ho gayi - trace me kya inspect karoge?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Embedding model migrate karna hai - rollout plan?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Prompt injection retrieved document ke andar mila - kya karoge?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Customer support agent ko refund action dena hai - permission design?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - AI analytics copilot wrong SQL generate karta hai - safety architecture?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Kafka lag grow kar raha hai - investigation order?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Duplicate streaming events aa rahe hain - design?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Late events badh gaye - watermark decision?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Backfill ke baad dashboard change ho gaya - impact analysis?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Model offline AUC improve hua but business metric flat - possible reasons?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Imbalanced classifier me threshold kaise tune?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - A/B test significant nahi - kya iska matlab no effect hai?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Feature leakage ka example do.

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk

boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Time-series model ka validation kaise?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Feature store ki need kya?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Data contract break hua to downstream blast radius kaise control?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Why lakehouse?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Why orchestration layer?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - LangChain vs LangGraph ka decision kaise?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - MCP use karne ki practical need?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Multi-agent ko single agent se compare kaise evaluate?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Memory me wrong fact store ho gaya - recovery?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par

depend karega.
Level: Scenario

Sawal - Agent ko max steps dena kyun?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Verifier ko same model se use karna risk hai?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - High-risk tool action par human approval kaise?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Prompt version rollback kaise?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - LLM evaluation me holdout set kyun?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - LLM-as-judge ki limitation?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Observability me trace/span difference kaise explain?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Cost per task kaise measure?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.
Level: Scenario

Sawal - Canary rollout kya protect karta hai?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Data drift aur model drift ka difference?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Data quality alert me false positives kaise reduce?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Incident ke baad postmortem me kya likhoge?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

Sawal - Apne project ki sabse important engineering trade-off kaunsi thi?

Jawaab - Pehle problem ko isolate karo, phir evidence collect karo, deterministic checks ko LLM reasoning se alag rakho, risk boundary define karo aur measured fix ke baad regression test chalao. Exact answer scenario ke data aur architecture par depend karega.

Level: Scenario

60-day mastery plan

Window	Focus	Deliverable
Day 1-7	**LLM + Transformer**	Tokenization, attention, embeddings, decoding; 35 interview answers aloud
Day 8-14	**RAG + Context**	Chunking, hybrid retrieval, reranking, ACL, eval set; build one grounded demo
Day 15-21	**LangChain + Tools**	Agent harness, tool contracts, structured output, middleware; build 3 tools
Day 22-28	**LangGraph + Memory**	State, branching, retry, persistence, HITL; build one workflow agent
Day 29-35	**MCP + Multi-Agent**	MCP server/client mental model, tool routing, supervisor pattern; build a safe tool server
Day 36-42	**Data Analysis**	Advanced SQL, cohorts, funnels, KPI, experiments, dashboards; solve 40 SQL/analytics problems
Day 43-49	**Data Science**	Stats, ML metrics, leakage, calibration, time-series, error analysis; solve 35 DS questions
Day 50-56	**Data Engineering**	CDC, Kafka, Spark, lakehouse, orchestration, data quality, lineage; design two pipelines
Day 57-60	**Production + Interview**	Security, observability, system design, project defense, 2 full mock interviews

Final master checklist

[] LLM internals + transformer intuition
[] Prompt/context engineering
[] Structured output
[] RAG + ACL + reranking
[] LangChain agents + tools
[] LangGraph stateful workflows
[] MCP tools/resources/prompts

[] Memory + HITL
[] Multi-agent coordination
[] Agent evaluation + termination
[] Data analysis + advanced SQL
[] Statistics + experiments
[] ML + model evaluation
[] Feature engineering + leakage
[] Time-series reasoning
[] Data warehouse + lakehouse
[] CDC + ETL/ELT
[] Kafka + streaming
[] Spark + state/watermarks
[] Orchestration + backfill
[] Data quality + lineage + contracts
[] MLOps + LLMOps
[] Security + governance
[] Observability + incident response
[] System design + scale + cost
[] Project defense + behavioral answers

Last interview rule

Answer ka structure rakho: Definition -> real example -> design decision -> trade-off -> failure case -> how you would measure success. Isi structure se tum basic answer ko advanced engineering answer me convert kar sakte ho.

28 SOURCES + 2026 VERIFICATION NOTES

Uploaded files primary base hain. Jahan 2026 framework/protocol details add kiye gaye hain, unhe current external verification ke roop me mark kiya gaya hai.

Uploaded source set

- LLMs.pdf - LLM ki buniyaad, history, tokenization, embeddings, Transformer, generation aur RAG concepts.
- AI Engineer Interview Q&A.pdf - LLM, RAG, production AI aur ML/DL interview foundations.
- LLM Interview Questions.pdf - tokenization, attention, context window, LoRA/QLoRA, decoding aur advanced LLM interview coverage.
- Algorithms Notes for Professionals - complexity, trees, graphs, BFS/DFS, Dijkstra, A*, DP, greedy, searching/sorting aur string algorithms.
- New Data Science with Generative AI - data/statistics, visualization, ML/DL, NLP, GenAI, Big Data, Spark/Kafka/Hadoop, capstones aur role pathways.
- AVO research paper - autonomous coding agent, tools, memory, evaluation aur iterative optimization loop.

Current external verification - 2026

- LangChain official JS docs - current create_agent harness, tools, middleware, model abstraction aur LangGraph integration: https://docs.langchain.com/oss/javascript/langchain/overview
- LangGraph official JS docs - stateful orchestration, persistence, human-in-the-loop, memory aur durable execution: https://docs.langchain.com/oss/javascript/langgraph/overview
- Deep Agents official docs - planning, subagents, filesystem, context management, memory, HITL, MCP aur long-running execution: https://docs.langchain.com/oss/javascript/deepagents/overview
- MCP 2026-07-28 official specification/release - stateless core, routing, caching aur authorization updates: https://modelcontextprotocol.io/specification/2026-07-28
- Apache Airflow official docs - DAG, tasks, dependencies, scheduling, retries aur backfill concepts: https://airflow.apache.org/docs/apache-airflow/stable/core-concepts/
- Apache Spark Structured Streaming guide: https://spark.apache.org/docs/latest/structured-streaming-programming-guide.html
- Delta Lake official docs - ACID, schema enforcement/evolution, time travel, streaming/batch aur merge operations: https://docs.delta.io/
- Apache Iceberg official docs - table evolution aur snapshot/time-travel concepts: https://iceberg.apache.org/docs/latest/
- Great Expectations official docs - Expectations, validation definitions aur production checkpoints: https://docs.greatexpectations.io/docs/
- Feast docs - offline/online stores aur point-in-time correct historical feature joins: https://docs.feast.dev/

Important scope note

Frameworks, model APIs aur protocol details change ho sakte hain. Is PDF me 2026 official docs se time-sensitive pieces verify kiye gaye hain; implementation se pehle exact provider/framework documentation aur package version check karo.